

AOS-CX 10.16xxxx IP Services Guide

4100i, 6000, 6100 Switch Series



**Hewlett Packard
Enterprise**

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Contents

About this document	8
Applicable products	8
Latest version available online	8
Command syntax notation conventions	8
About the examples	9
Identifying switch ports and interfaces	9
IRDP	11
Configuring IRDP	12
IRDP commands	13
diag-dump irdp basic	13
ip irdp	14
ip irdp holdtime	15
ip irdp maxadvertinterval	16
ip irdp minadvertinterval	17
ip irdp preference	17
show ip irdp	18
IPv6 Router Advertisement	20
Configuring IPv6 RA	20
IPv6 RA scenario	22
IPv6 RA commands	22
ipv6 address <global-unicast-address>	22
ipv6 address autoconfig	23
ipv6 address link-local	24
ipv6 nd cache-limit	25
ipv6 nd dad attempts	25
ipv6 nd hop-limit	26
ipv6 nd mtu	27
ipv6 nd ns-interval	27
ipv6 nd prefix	28
ipv6 nd ra dns search-list	30
ipv6 nd ra dns server	30
ipv6 nd ra lifetime	31
ipv6 nd ra managed-config-flag	32
ipv6 nd ra max-interval	33
ipv6 nd ra min-interval	34
ipv6 nd ra other-config-flag	35
ipv6 nd ra reachable-time	36
ipv6 nd ra retrans-timer	36
ipv6 nd route	37
ipv6 nd router-preference	38
ipv6 nd suppress-ra	39
show ipv6 nd global traffic	39
show ipv6 nd interface	40
show ipv6 nd interface prefix	45
show ipv6 nd interface route	46
show ipv6 nd ra dns search-list	47
show ipv6 nd ra dns server	48

sFlow	49
sFlow agent	49
Configuring the sFlow agent	50
sFlow scenario	51
sFlow scenario 2	52
sFlow agent commands	54
clear sflow statistics	54
sflow	55
sflow agent-ip	56
sflow collector	57
sflow disable	58
sflow header-size	58
sflow max-datagram-size	59
sflow polling	60
sflow sampling	60
show sflow	61
DHCP	64
DHCP client	64
Protocol and feature details	64
Supported platform and standards	64
Configuration task list	65
Considerations and best practices	65
Use case	65
DHCP client commands	67
ip dhcp	67
ipv6 dhcp	68
ip dhcp option	69
show ip dhcp	70
show ipv6 dhcp	71
DHCP relay agent	71
Supported platform and standards	71
Protocol and feature details	72
DHCPv4 relay agent	72
Configuring the DHCPv4 relay agent	73
Use Case	74
DHCPv4 relay commands	77
DHCPv6 relay agent	85
Configuring the DHCPv6 relay agent	86
Use Case	87
DHCP relay (IPv6) commands	88
FAQ	93
DHCP snooping	96
Protocol details	97
Supported platform and standards	97
DHCPv4 Snooping Use case	97
DHCP snooping commands	100
clear dhcp-snooping binding	100
clear dhcp-snooping statistics	101
dhcp-snooping	102
dhcp-snooping (in config-vlan context)	103
dhcp-snooping allow-overwrite-binding	104
dhcp-snooping authorized-server	105
dhcp-snooping event-log client	106
dhcp-snooping external-storage	107

dhcp-snooping flash-storage	108
dhcp-snooping max-bindings	109
dhcp-snooping option 82	110
dhcp-snooping static-attributes	112
dhcp-snooping trust	113
dhcp-snooping verify mac	113
show dhcp-snooping	114
show dhcp-snooping binding	116
show dhcp-snooping statistics	117
DHCPv6 snooping commands	118
clear dhcpv6-snooping binding	118
clear dhcpv6-snooping statistics	119
dhcpv6-snooping	120
dhcpv6-snooping (in config-vlan context)	121
dhcpv6-snooping authorized-server	121
dhcpv6-snooping event-log client	123
dhcpv6-snooping external-storage	123
dhcpv6-snooping flash-storage	125
dhcpv6-snooping max-bindings	126
dhcpv6-snooping trust	127
show dhcpv6-snooping	128
show dhcpv6-snooping binding	129
show dhcpv6-snooping statistics	130
Troubleshooting	131
DHCP Options	133
DHCP options commands	133
http-proxy	133
IP Source Lockdown	136
IPv4 source lockdown commands	136
ipv4 source-binding	136
ipv4 source-lockdown	137
ipv4 source-lockdown hardware retry	138
show ipv4 source-binding	139
show ipv4 source-lockdown	140
IPv6 source lockdown commands	143
ipv6 source-binding	143
ipv6 source-lockdown	144
ipv6 source-lockdown hardware retry	145
show ipv6 source-binding	146
show ipv6 source-lockdown	147
Internet Control Message Protocol (ICMP)	151
ICMP message types	151
When ICMP messages are sent	151
ICMP redirect messages	152
When ICMP redirect messages are sent	152
ICMP commands	152
ip icmp redirect	152
ip icmp throttle	153
ip icmp unreachable	154
DNS	155
DNS client	155
Configuring the DNS client	155

DNS client commands	156
ip dns domain-list	156
ip dns domain-name	157
ip dns host	157
ip dns server address	158
show ip dns	159
ARP	162
ARP commands	165
arp inspection	165
arp inspection trust	166
arp ip	166
arp process-grat-arp	167
clear arp	168
debug arp-security	169
ip proxy-arp	171
ipv6 neighbor mac	172
show arp	173
show arp inspection interface	174
show arp inspection statistics	175
show arp inspection vlan	176
show arp state	177
show arp summary	178
show arp timeout	179
show arp vrf	180
show ipv6 neighbors	181
show ipv6 neighbors state	182
show tech arp-security	183
Support and Other Resources	185
Accessing HPE Aruba Networking Support	185
Accessing Updates	186
Warranty Information	186
Regulatory Information	186
Documentation Feedback	186

Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <i>example-text</i> ■ <code>example-text</code> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (<code>< ></code>). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for

Convention	Usage
	readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

ICMP Router Discovery Protocol (IRDP), an extension of the ICMP, is independent of any routing protocol. It allows hosts to discover the IP addresses of neighboring routers that can act as default gateways to reach devices on other IP networks.

On the switches covered by this guide, IRDP is configured on a VLAN interface.

IRDP operation

IRDP uses the following types of ICMP messages:

- Router advertisement (RA): Sent by a router to advertise IP addresses (including the primary and secondary IP addresses) and preference.
- Router solicitation (RS): Sent by a host to request the IP addresses of routers on the subnet.

An interface with IRDP enabled periodically broadcasts or multicasts an RA message to advertise its IP addresses. A receiving host adds the IP addresses to its routing table, and selects the IP address with the highest preference as the default gateway.

When a host attached to the subnet starts up, the host multicasts an RS message to request immediate advertisements. If the host does not receive any advertisements, it retransmits the RS several times. If the host does not discover the IP addresses of neighboring routers because of network problems, the host can still discover them from periodic RAs.

IRDP allows hosts to discover neighboring routers, but it does not suggest the best route to a destination. If a host sends a packet to a router that is not the best next hop, the host will receive an ICMP redirect message from the router.

IP address preference

Every IP address advertised in RAs has a preference value. A larger preference value represents a higher preference. The IP address with the highest preference is selected as the default gateway address.

You can specify the preference for IP addresses to be advertised on a router interface.

An address with the minimum preference value (-2147483648) will not be used as a default gateway address.

Lifetime of an IP address

An RA contains a lifetime field that specifies the lifetime of advertised IP addresses. If the host does not receive a new RA for an IP address within the address lifetime, the host removes the route entry.

All the IP addresses advertised by an interface have the same lifetime.

Advertising interval

A router interface with IRDP enabled sends out RAs randomly between the minimum and maximum advertising intervals. This mechanism prevents the local link from being overloaded by a large number of RAs sent simultaneously from routers.

As a best practice, shorten the advertising interval on a link that suffers high packet loss rates

Destination address of RA

An RA uses either of the following destination IP addresses:

- Broadcast address 255.255.255.255.
- Multicast address 224.0.0.1, which identifies all hosts on the local link.

By default, the destination IP address of an RA is the multicast address. If all listening hosts in a local area network support IP multicast, specify 224.0.0.1 as the destination IP address.

Proxy-advertised IP addresses

By default, an interface advertises its primary and secondary IP addresses. You can specify IP addresses of other gateways for an interface to proxy-advertise.

VRF support

In IP-based computer networks, virtual routing and forwarding (VRF) is a technology that allows multiple instances of a routing table to co-exist within the same router at the same time. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other.

IRDP is VRF aware. As the router advertisements and solicit processing occurs on the interface, packet is through the interface and corresponding VRF.

VSX synchronization

IRDP supports VSX synchronization. For more information on using VSX, see the *Virtual Switching Extension (VSX) Guide* for your switch and software version

Configuring IRDP

Prerequisites

A layer 3 interface.

Procedure

1. Enable IRDP on an interface with the command `ip irdp`.
2. Set the maximum hold time with the command `ip irdp holdtime`.
3. Set the maximum router advertisement interval with the command `ip irdp maxadvertinterval`.
4. Set the minimum router advertisement interval with the command `ip irdp minadvertinterval`.
5. Set the IRDP preference level with the command `ip irdp preference`.
6. Review IRDP configuration settings with the command `show ip irdp`.

Example

This example creates the following configuration:

- Enables IRDP on the layer 3VLAN interface 2 with packet type set to broadcast.
- Sets the hold time to 5000 seconds.
- Sets the advertisement interval to 30 seconds.
- Sets the minimum advertisement interval to 25 seconds.
- Sets the IRDP preference level to 25.

```

switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
switch(config-if-vlan)# ip irdp holdtime 5000
switch(config-if-vlan)# ip irdp maxadvertinterval 30
switch(config-if-vlan)# ip irdp minadvertinterval 25
switch(config-if-vlan)# ip irdp preference 25

```

IRDP commands

diag-dump irdp basic

diag-dump irdp basic

Description

Displays diagnostic information for IRDP.

Example

```

switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jun  8 09:50:28 2017
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: 1/1/1 (state : Up)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.1.2,
Interface: 1/1/2 (state : Up)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.2.2,
-----
[End] Daemon hpe-rdiscd
-----
=====
[End] Feature irdp
=====
Diagnostic dump captured for feature irdp

```

```

switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jan  7 04:46:25 2021
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: vlan2 (state : Down)
rdisc ipv4 (enabled: 1, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
Interface: vlan1 (state : Down)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
-----

```

```
[End] Daemon hpe-rdiscd
-----
[End] Feature irdp
-----
Diagnostic-dump captured for feature irdp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	

ip irdp

```
ip irdp [broadcast | multicast]
no ip irdp
```

Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces. The **no** form of this command disables IRDP on an interface.

Parameter	Description
<code>broadcast</code>	Advertisements are sent as broadcast packets to IP address 255.255.255.255.
<code>multicast</code>	Advertisements are sent as multicast packets to the multicast group with IP address 24.0.0.1. Default.

Examples

Enabling IRDP on interface vlan 2 with packet type set to the default value (multicast).

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp
```

Enabling IRDP on interface vlan 2 with packet type set to broadcast.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp holdtime

```
ip irdp holdtime <TIME>
no ip irdp holdtime <TIME>
```

Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

The **no** form of this command removes the specified maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives and update it to the default value.

Parameter	Description
<TIME>	Specifies the lifetime of router advertisements sent from this interface. Range: 4 to 9000 seconds. Default: 1800 seconds.

Example

Setting the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp holdtime 5000
```

Removing the the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp holdtime 5000
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp maxadvertinterval

```
ip irdp maxadvertinterval <TIME>
no ip irdp maxadvertinterval <TIME>
```

Description

Specifies the maximum router advertisement interval.

The **no** form of this command removes the specified maximum router advertisement interval and reverts to the default value.

Parameter	Description
<TIME>	Specifies the maximum time allowed between the sending of unsolicited router advertisements. Range: 4 to 1800 seconds. Default: 600 seconds.

Example

Setting the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp maxadvertinterval 30
```

Removing the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp maxadvertinterval 30
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp minadvertinterval

```
ip irdp minadvertinterval <TIME>
no ip irdp minadvertinterval <TIME>
```

Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.

The **no** form of this command removes the specified minimum amount of time the switch waits between sending router advertisements and reverts to the default value.

Parameter	Description
<TIME>	Specifies the minimum time allowed between the sending of unsolicited router advertisements. Range: 3 to 1800 seconds. Default: 450 seconds (75% of the default value for maximum router advertisement interval).

Example

Setting the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp minadvertinterval 25
```

Removing the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp minadvertinterval 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp preference

```
ip irdp preference <LEVEL>
no ip irdp preference <LEVEL>
```

Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

The **no** form of this command removes the specified IRDP preference level and reverts to the default value.

Parameter	Description
<LEVEL>	Specifies the IRDP preference level. Range: -2147483648 to 2147483647. Default: 0.

Example

Setting the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp preference 25
```

Removing the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp preference 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

show ip irdp

```
show ip irdp
```

Description

Displays IRDP configuration settings.

Example

```
switch# show ip irdp

ICMP Router Discovery Protocol

Interface Status   Advertising Minimum   Maximum   Holdtime Preference
                  Address      Interval  Interval

```

```

-----
1/1/1      Enabled  multicast  6          8          10         10
1/1/2      Disabled multicast  450        600        1800       0
1/1/3      Enabled  broadcast  450        600        1800      115
-----

```

```
switch# sh ip irdp
```

```
ICMP Router Discovery Protocol
```

```

Interface      Status   Advertising Address  Minimum  Maximum  Holdtime  Preference
-----
vlan1          Disabled multicast  450      600      1800      0
bridge_normal  Disabled multicast  450      600      1800      0

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 3

IPv6 Router Advertisement

IPv6 RA provides a method for local IPv6 hosts to automatically configure their own IP address (and other settings such as a preferred DNS server) based on information advertised by switches/routers operating on the network.

IPv6 flags

Behavior of IPv6 hosts to IPv6 RA messages is controlled by the managed address configuration flag (M flag), and other stateful configuration flag (O flag).

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Configuring IPv6 RA

Procedure

1. Enable transmission of IPv6 router advertisements with the command `no ipv6 nd suppress-ra`.
2. Optionally, configure IPv6 unicast address prefixes with the command `ipv6 nd prefix`.
3. Optionally, configure support for DNS name resolution with the commands `ipv6 nd ra dns server` and `ipv6 nd ra dns search-list`.
4. For most deployments, the default values for the following features do not need to be changed. If your deployment requires different settings, change the default values with the indicated command:

IPv6 RA setting	Default value	Command to change it
Number of neighbor solicitations to be sent when performing DAD.	1	<code>ipv6 nd dad attempts</code>
Number of neighbor entries in the ND cache.	131072	<code>ipv6 nd cache-limit</code>

IPv6 RA setting	Default value	Command to change it
Hop limit to be sent in the RA messages.	64	<code>ipv6 nd hop-limit</code>
MTU value to be sent in the RA messages.	1500 bytes	<code>ipv6 nd mtu</code>
Neighbor solicitation interval	1000 milliseconds	<code>ipv6 nd ns-interval</code>
Lifetime of a default router.	1800 seconds	<code>ipv6 nd ra lifetime</code>
Retrieval of an IPv6 address by devices.	Disabled	<code>ipv6 nd ra managed-config-flag</code>
Maximum interval between transmissions of IPv6 RAs.	600 seconds	<code>ipv6 nd ra max-interval</code>
Minimum interval between transmissions of IPv6 RAs.	200 seconds	<code>ipv6 nd ra min-interval</code>
Time that an interface considers a device to be reachable.	0 milliseconds (no limit)	<code>ipv6 nd ra reachable-time</code>
Retry period between ND solicitations.	0 (Use locally configured NS-interval)	<code>ipv6 nd ra retrans-timer</code>
Default routing preference for an interface.	Medium	<code>ipv6 nd router-preference</code>

5. Review IPv6 RA configuration settings with the commands `show ipv6 nd interface`, `show ipv6 nd interface prefix`, `show ipv6 nd ra dns server`, and `show ipv6 nd ra dns search-list`.

Example

This example creates the following configuration:

- Enables IPV6 RA on interface `interface vlan 2`.
- Sets the recursive DNS server address to `4001::1` with a lifetime of 400 seconds.
- Sets the minimum interval between transmissions to 3 seconds.
- Sets the maximum interval between transmissions to 13 seconds.
- Sets the lifetime of a default router to 1900 seconds.

```
switch(config)# interface vlan 2
switch(config-if)# no ipv6 nd suppress-ra
switch(config-if)# ipv6 nd ra dns server 4001::1 lifetime 400
switch(config-if)# ipv6 nd ra min-interval 3
switch(config-if)# ipv6 nd ra max-interval 13
switch(config-if)# ipv6 nd ra lifetime 1900
switch(config-if)# end
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
Admin state is up
IPv6 address:
  2006::1/64 [VALID]
IPv6 link-local address: fe80::98f2:b321:368:6dc6/64 [VALID]
ICMPv6 active timers:
```

```

    Last Router-Advertisement sent: 0 Secs
    Next Router-Advertisement sent in: 13 Secs
Router-Advertisement parameters:
  Periodic interval: 3 to 13 secs
  Router Preference: medium
  Send "Managed Address Configuration" flag: false
  Send "Other Stateful Configuration" flag: false
  Send "Current Hop Limit" field: 64
  Send "MTU" option value: 1500
  Send "Router Lifetime" field: 1900
  Send "Reachable Time" field: 0
  Send "Retrans Timer" field: 0
  Suppress RA: false
  Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: vlan2
  Suppress DNS Server List: No
  DNS Server 1: 2001::1    lifetime 400

```

IPv6 RA scenario

In this scenario, two host computers are auto-configured with IP addresses using IPv6 RA. In addition, the switch provides the hosts with an address of a recursive DNS server.

Procedure

1. Configure the VLAN interfaces with IPv6 addresses.


```

switch# config
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 2001::1/64
switch(config)# interface vlan 3
switch(config-if-vlan)# ipv6 address 3001::1/64
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 address 4001::1/64

```
2. Enable transmission of all IPv6 RA messages.


```

switch(config-if-vlan)# no ipv6 nd suppress-ra

```

IPv6 RA commands

ipv6 address <global-unicast-address>

```

ipv6 address <global-unicast-address>
no ipv6 address <global-unicast-address>

```

Description

Sets a global unicast address on the interface.

The **no** form of this command removes the global unicast address on the interface.

This command automatically creates an IPv6 link-local address on the interface. However, it does not add the **ipv6 address link-local command** to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the **ipv6 address link-local command**.

Example

Enabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 3731:54:65fe:2::a7
```

Disabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address 3731:54:65fe:2::a7
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	

ipv6 address autoconfig

```
ipv6 address autoconfig
no ipv6 address autoconfig
```

Description

Enables the interface to automatically obtain an IPv6 address using router advertisement information and the EUI-64 identifier.

The **no** form of this command disables address auto-configuration.

-
- A maximum of 15 autoconfigured addresses are supported.
 - This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.
-

Usage

The IPv6 SLAAC feature lets the router obtain the IPv6 address for the interface it is configured through the SLAAC method. This feature is not available on the `mgmt` VRF.

Example

Enabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address autoconfig
```

Disabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address autoconfig
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	

ipv6 address link-local

ipv6 address link-local [<IPV6-ADDR>/<MASK>]

Description

Enables IPv6 on the current interface. If no address is specified, an IPv6 link-local address is auto-generated for the interface. If an address is specified, auto-configuration is disabled and the specified address/mask is assigned to the interface.

To disable IPv6 link-local on the interface, remove **ipv6 address link-local**, **ipv6 address <global-ipv6-address>**, and **ipv6 address autoconfig** from the interface.

This feature is not available on the management VRF.

Parameter	Description
<IPV6-ADDR>	Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55 .
<MASK>	Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Example

Enabling IPv6 link-local on the interface:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address link-local
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	

ipv6 nd cache-limit

```
ipv6 nd cache-limit <CACHELIMIT>
no ipv6 nd cache-limit [<CACHELIMIT>]
```

Description

Configures the limit on the number of neighbor entries in the ND cache.
The **no** form of this command sets the cache limit to the default value.

Parameter	Description
<CACHELIMIT>	Specifies the neighbor cache entries limit. Range: 1-131072. Default: 131072.

Examples

Setting the cache limit to 20.

```
switch(config)# ipv6 nd cache-limit 20
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ipv6 nd dad attempts

```
ipv6 nd dad attempts <NUM-ATTEMPTS>
no ipv6 nd dad attempts [<NUM-ATTEMPTS>]
```

Description

Configures the number of neighbor solicitations to be sent when performing duplicate address detection (DAD) for a unicast address configured on an interface. If the active gateway is configured with the same IP as an SVI IP, then IPv6 DAD cannot be configured.

The **no** form of this command sets the number of attempts to the default value.

Parameter	Description
<code>dad attempts <NUM-ATTEMPTS></code>	Specifies the number of neighbor solicitations to send. Range: 0-15. Default: 1.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd dad attempts 5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if-vlan</code>	Administrators or local user group members with execution rights for this command.

ipv6 nd hop-limit

```
ipv6 nd hop-limit <HOPLIMIT>
no ipv6 nd hop-limit [<HOPLIMIT>]
```

Description

Configures the hop limit to be sent in RAs.

The **no** form of this command resets the hop limit to 0. This reset eliminates the hop limit from the RAs that originate on the interface, so the host determines the hop limit.

Parameter	Description
<code>hop-limit <HOPLIMIT></code>	Specifies the hop limit. Range: 0-255. Default: 64.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd hop-limit 64
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd mtu

```
ipv6 nd mtu <MTU-VALUE>
no ipv6 nd mtu [<MTU-VALUE>]
```

Description

Configures the MTU size to be sent in the RA messages.

The **no** form of this command sets hop limit to the default value.

Parameter	Description
<MTU-VALUE>	Specifies the MTU size. Range: 1280-65535 bytes. Default: 1500 bytes.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd mtu 1300
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ns-interval

```
ipv6 nd ns-interval <TIME>
```

```
no ipv6 nd ns-interval [<TIME>]
```

Description

Configures the ND time in milliseconds between DAD neighbor solicitations sent for an unresolved destination. Increase the ns-interval time if the network is slow or if there are persistent retry failures. If the active gateway is configured with the same IP as an SVI IP, then IPv6 DAD cannot be configured

The **no** form of this command sets the ns-interval to the default value.

Parameter	Description
<TIME>	Specifies the neighbor solicitation interval. Range: 1000-3600000 milliseconds. Default: 1000 milliseconds.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ns-interval 1200
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd prefix

```
ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN>
    [no-advertise | [valid <LIFETIME-VALUE> preferred
    <LIFETIME-VALUE>] | no-autoconfig | no-onlink]

no ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN> [no-advertise
    | [valid <LIFETIME-VALUE> preferred <LIFETIME-VALUE>
    ] | no-autoconfig | no-onlink]

ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}

no ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}
```

Description

Specifies prefixes for the routing switch to include in RAs transmitted on the interface. IPv6 hosts use the prefixes in RAs to autoconfigure themselves with global unicast addresses. The autoconfigured address of a host is composed of the advertised prefix and the interface identifier in the current link-local address of the host.

By default, advertise, autoconfig, and onlink are set.

The **no** form of this command removes the configuration on the interface.

Parameter	Description
<code><IPv6-ADDR>/<PREFIX-LEN></code>	Specifies the IPv6 prefix to advertise in RA. Format: X:X::X/M
<code>default</code>	Specifies apply configuration to all on-link prefixes that are not individually set by the ipv6 ra prefix <IPv6-ADDR>/<PREFIX-LEN> command. It applies the same valid and preferred lifetimes, link state, autoconfiguration state, and advertise options to the advertisements sent for all on-link prefixes that are not individually configured with a unique lifetime. This also applies to the prefixes for any global unicast addresses configured later on the same interface. Using default once, and then using it again with any new parameter values results in the new values replacing the former values in advertisements. If default is used without the no-advertise, no-autoconfig, or no-onlink parameter, the advertisement setting for the absent parameter is returned to its default setting.
<code>no-advertise</code>	Specifies do not advertise prefix in RA.
<code>valid <LIFETIME-VALUE></code>	Specifies the total time, in seconds, the prefix remains available before becoming unusable. After preferred-lifetime expiration, any autoconfigured address is deprecated and used only for transactions only before preferred-lifetime expires. If the valid lifetime expires, the address becomes invalid. You can enter a value in seconds or enter valid infinite which sets infinite lifetime. Default: 2,592,000 seconds which is 30 days. Range: 0-4294967294 seconds.
<code>preferred <LIFETIME-VALUE></code>	Specifies the span of time during which the address can be freely used as a source and destination for traffic. This setting must be less than or equal to the corresponding valid-lifetime setting. You can enter a value in seconds or enter preferred infinite which sets infinite lifetime. Default: 604,800 seconds which is seven days. Range: 0-4294967294 seconds.
<code>no-autoconfig</code>	Specifies do not use prefix for autoconfiguration.
<code>no-onlink</code>	Specifies do not use prefix for onlink determination.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd prefix 4001::1/64 valid 30 preferred 10 no-
autoconfig no-onlink
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra dns search-list

```
ipv6 nd ra dns search-list <DOMAIN-NAME> [lifetime <TIME>]
no ipv6 nd ra dns search-list <DOMAIN-NAME>
```

Description

Configures the DNS Search List (DNSSL) to include in Router Advertisements (RAs) transmitted on the interface.

The **no** form of this command removes the DNS Search List from the RAs transmitted on the interface.

Parameter	Description
<DOMAIN-NAME>	Specifies the domain names for DNS queries.
lifetime <TIME>	Specifies lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Usage

- DNSSL contains the domain names of DNS suffixes or IPv6 hosts to append to short, unqualified domain names for DNS queries.
- Multiple DNS domain names can be added to the DNSSL by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com lifetime 500
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra dns server

```
ipv6 nd ra dns server <IPV6-ADDR> [lifetime <TIME>]
no ipv6 nd ra dns server <IPV6-ADDR>
```

Description

Configures the IPv6 address of a preferred Recursive DNS Server (RDNSS) to be included in Router Advertisements (RAs) transmitted on the interface.

The **no** form of this command removes the configured DNS server from the RAs transmitted on the interface.

Parameter	Description
<code><IPv6-ADDR></code>	Specifies the RDNSS address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55 .
<code>lifetime <TIME></code>	Specifies IPv6 DNS server lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Usage

- Including RDNSS information in RAs provides DNS server configuration for connected IPv6 hosts without requiring DHCPv6.
- Multiple servers can be configured on the interface by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1 lifetime 400
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra lifetime

```
ipv6 nd ra lifetime <TIME>
no ipv6 nd ra lifetime [<TIME>]
```

Description

Configures the lifetime, in seconds, for the routing switch to be used as a default router by hosts on the current interface.

The **no** form of this command sets lifetime to the default of 1800 seconds.

Parameter	Description
<code><TIME></code>	Specifies lifetime in seconds of a default router. A setting of 0 for default router lifetime in an RA indicates that the routing switch is not a default router on the interface. Range: 0-9000 seconds. Default: 1800 seconds.

Usage

- A given host on an interface refreshes the default router lifetime for a specific router each time the host receives an RA from that router.
- A specific router ceases to be a default router candidate for a given host if the default router lifetime expires before the host is updated with a new RA from the router.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra lifetime 1200
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra managed-config-flag

```
ipv6 nd ra managed-config-flag
no ipv6 nd ra managed-config-flag
```

Description

Controls the M flag setting in RAs the router transmits on the current interface. Enable the M flag to indicate that hosts can obtain IP address through DHCPv6. The M flag is disabled by default.

The **no** form of this command turns off (disables) the M flag.

Usage

- Enabling the M flag directs hosts to acquire their IPv6 addressing for the current interface from a DHCPv6 server.
- When the M-bit is enabled, receiving hosts ignore the O flag setting, which is configured using the command **ipv6 nd ra other-config-flag**.

- When the M-bit is disabled (the default), receiving hosts expect to receive their IPv6 addresses from RA.

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra managed-config-flag
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra max-interval

```
ipv6 nd ra max-interval <TIME>
no ipv6 nd ra max-interval [<TIME>]
```

Description

Configures the maximum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The **no** form of this command returns the setting to its default, provided the default value is less than the default lifetime value.

Parameter	Description
<TIME>	Specifies the maximum advertisement time in seconds. Range: 4-1800. Default: 600 seconds.

Usage

- This value has one setting per interface. The setting does not apply to RAs sent in response to a router solicitation received from another device.
- Attempting to set max-interval to a value that is not sufficiently larger than the current min-interval also results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra max-interval 30
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra min-interval

```
ipv6 nd ra min-interval <TIME>
no ipv6 nd ra min-interval [<TIME>]
```

Description

Configures the minimum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The **no** form of this command returns the setting to its default, provided the default value is less than the current max-interval setting.

Parameter	Description
<TIME>	Specifies a minimum advertisement time in seconds. Range: 3-1350. Default: 200 seconds.

Usage

- This value has one setting per interface and does not apply to RAs sent in response to a router solicitation received from another device.

- The min-interval must be less than the max-interval. Attempting to set min-interval to a higher value results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra min-interval 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra other-config-flag

```
ipv6 nd ra other-config-flag
no ipv6 nd ra other-config-flag
```

Description

Controls the O-bit in RAs the router transmits on the current interface; but is ignored unless the M-bit is disabled in RAs. Configure to set the O-bit in RA messages for host to obtain network parameters through DHCPv6. The other-config-flag is disabled by default.

For more information on configuring the M-bit, see **ipv6 nd ra managed-config-flag**.

The **no** form of this command turns off (disables) the setting for this command in RAs.

Usage

Enabling the O-bit while the M-bit is disabled directs hosts on the interface to acquire their other configuration information from DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra other-config-flag
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra reachable-time

```
ipv6 nd ra reachable-time <TIME>
no ipv6 nd ra reachable-time [<TIME>]
```

Description

Sets the amount of time that the interface considers a device to be reachable after receiving a reachability confirmation from the device.

The **no** form of this command sets the reachable time to the default value of 0. (no limit).

Parameter	Description
<TIME>	Specifies the reachable time in milliseconds. Range: 1000-3600000. Default: 0 (no limit).

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra reachable-time 2000
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra retrans-timer

```
ipv6 nd ra retrans-timer <TIME>
no ipv6 nd ra retrans-timer [<TIME>]
```

Description

Configures the period (retransmit timer) between ND solicitations sent by a host for an unresolved destination, or between DAD neighbor solicitation requests. By default, hosts on the interface use their own locally configured NS-interval settings instead of using the value received in the RAs.

Increase this timer when neighbor solicitation retries or failures are occur, or in a "slow" (WAN) network.

The **no** form of this command sets the value to the default of 0.

Parameter	Description
<TIME>	Specifies the retransmit timer value in milliseconds. Range: 0 - 4294967295 milliseconds. Default: 0 (Use locally configured NS-interval).

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra retrans-timer 400
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd route

```
ipv6 nd route <IPV6-ADDR>/<PREFIX-LEN> [no-advertise | lifetime {<SECONDS> | infinite} |
preference {low | medium | high}]
no ipv6 nd route <IPV6-ADDR>/<PREFIX-LEN> [no-advertise | lifetime {<SECONDS> | infinite}
| preference {low | medium | high}]
```

Description

Configures the routing switch to include the routing information in the RAs transmitted on the interface. The routing switch includes the route information in the RA packets only if the configured routes are present in the routing table. After receiving the RA packets carrying the route information, the IPv6 host updates its routing table. The hosts lookup their routing table and selects the best possible route to forward packets.

The **no** form of this command removes the settings for including the routing information in the RA packets.

Parameter	Description
<IPV6-ADDR>/<PREFIX-LEN>	Specifies the IPv6 route prefix to advertise in RA. Format: X:X::X:X/M
no-advertise	Specifies to not advertise the route information.
lifetime {<SECONDS> infinite}	Specifies the duration in seconds that the route is valid for the route determination. If this parameter is configured with 0 , the route becomes invalid. Default: 1800 . Range: 0-4294967295 .

Parameter	Description
preference {low medium high}	Specifies the preference for the hosts to choose the router associated with the route over other routers when multiple identical route prefixes from different routers are received. Default: medium

Examples

Configuring routing information on interface **1/1/1**.

```
switch(config)# int 1/1/1
switch(config-if)# ipv6 nd route 1::1/64 lifetime 200 preference high
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms		Administrators or local user group members with execution rights for this command.

ipv6 nd router-preference

```
ipv6 nd router-preference {high | medium | low}
no ipv6 nd router-preference [high | medium | low]
```

Description

Specifies the value that is set in the Default Router Preference (DRP) field of Router Advertisements (RAs) that the switch sends from an interface. An interface with a DRP value of high will be preferred by other devices on the network over interfaces with an RA value of medium or low.

The **no** form of this command set the value to the default of medium.

Parameter	Description
high	Sets DRP to high.
medium	Sets DRP to medium. Default.
low	Sets DRP to low.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd router-preference high
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd suppress-ra

```
ipv6 nd suppress-ra [<SUPPRESS-OPTION>]
no ipv6 nd ra supress-ra [<SUPPRESS-OPTION>]
```

Description

Configures suppression of IPv6 Router Advertisement transmissions on an interface. The **no** form of this command restores transmission of IPv6 Router Advertisement and options.

Parameter	Description
suppress-ra [<SUPPRESS-OPTION>]	Specifies suppressing RA transmissions. Entering suppress-ra without any options, suppresses all RA messages (default). Or you can enter one of the following options.
dnssl	Specifies suppressing DNSSL options in RA messages.
mtu	Specifies suppressing MTU options in RA messages.
rdnss	Specifies suppressing RDNSS options in RA messages.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd suppress-ra mtu dnssl rdnss
switch(config-if-vlan)# no ipv6 nd suppress-ra mtu dnssl rdnss
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

show ipv6 nd global traffic

```
show ipv6 nd global traffic
```

Description

Displays IPV6 Neighbor Discovery traffic details on a device.

Examples

```
switch# show ipv6 nd global traffic
ICMPv6 packet Statistics (sent/received)
  Total Messages           :      18/0
  Error Messages           :       0/0
  Destination Unreachables :       0/0
  Time Exceeded            :       0/0
  Parameter Problems       :       0/0
  Echo Request             :       0/0
  Echo Replies             :       0/0
  Redirects                :       0/0
  Packet Too Big           :       0/0
  Router Advertisements    :       4/0
  Router Solicitations     :       0/0
  Neighbor Advertisements  :       0/0
  Neighbor Solicitations   :       3/0
  Duplicate router RA received :    0/0
ICMPv6 MLD Statistics (sent/received)
  V1 Queries :       0/0
  V2 Queries :       0/0
  V1 Reports  :       0/0
  V2 Reports  :      11/0
  V1 Leaves  :       0/0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 nd interface

```
show ipv6 nd interface [<IF-NAME> | all-vrfs | vrf <VRF-NAME>]
```

Description

Displays neighbor discovery information for an interface. If no options are specified, displays information for the default VRF.

Parameter	Description
<IF-NAME>	Displays information about the specified IPv6 enabled interface.

Parameter	Description
all-vrfs	Displays information about interfaces in all VRFs.
vrf <VRF-NAME>	Displays information about interfaces in a particular VRF. Or, if <VRF-NAME> is not specified, information for the default VRF is displayed.

Examples

Showing information for all VRFs:

```
switch# show ipv6 nd interface all-vrfs

List of IPv6 Interfaces for VRF default
Interface 1/1/1 is up
  Admin state is up
  IPv6 address:
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1

List of IPv6 Interfaces for VRF red
Interface 1/1/2 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
```

```
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
```

```
switch# show ipv6 nd interface all-vrfs
```

```
List of IPv6 Interfaces for VRF default
```

```
Interface vlan2 is up
```

```
Admin state is up
```

```
IPv6 address:
```

```
IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
```

```
ICMPv6 active timers:
```

```
  Last Router-Advertisement sent:
```

```
  Next Router-Advertisement sent in:
```

```
Router-Advertisement parameters:
```

```
  Periodic interval: 200 to 600 secs
```

```
  Router Preference: medium
```

```
  Send "Managed Address Configuration" flag: false
```

```
  Send "Other Stateful Configuration" flag: false
```

```
  Send "Current Hop Limit" field: 64
```

```
  Send "MTU" option value: 1500
```

```
  Send "Router Lifetime" field: 1800
```

```
  Send "Reachable Time" field: 0
```

```
  Send "Retrans Timer" field: 0
```

```
  Suppress RA: true
```

```
  Suppress MTU in RA: true
```

```
ICMPv6 error message parameters:
```

```
  Send redirects: false
```

```
ICMPv6 DAD parameters:
```

```
  Current DAD attempt: 1
```

```
List of IPv6 Interfaces for VRF red
```

```
Interface vlan3 is up
```

```
Admin state is up
```

```
IPv6 address:
```

```
  2001::1/64 [VALID]
```

```
IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
```

```
ICMPv6 active timers:
```

```
  Last Router-Advertisement sent:
```

```
  Next Router-Advertisement sent in:
```

```
Router-Advertisement parameters:
```

```
  Periodic interval: 200 to 600 secs
```

```
  Router Preference: medium
```

```
  Send "Managed Address Configuration" flag: false
```

```
  Send "Other Stateful Configuration" flag: false
```

```
  Send "Current Hop Limit" field: 64
```

```
  Send "MTU" option value: 1500
```

```
  Send "Router Lifetime" field: 1800
```

```
  Send "Reachable Time" field: 0
```

```
  Send "Retrans Timer" field: 0
```

```
  Suppress RA: true
```

```
  Suppress MTU in RA: true
```

```
ICMPv6 error message parameters:
```

```
  Send redirects: false
```

```
ICMPv6 DAD parameters:
```

```
  Current DAD attempt: 1
```

Showing information for interface vlan 2:

```

switch# show ipv6 nd interface 1/1/1
Interface 1/1/1 is up
  Admin state is up
  IPv6 address:
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: high
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1

```

```

switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
  Admin state is up
  IPv6 address:
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: high
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1

```

Showing information for the default VRF:

```

switch# show ipv6 nd interface

List of IPv6 Interfaces for VRF default
Interface 1/1/1 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]

```

```
IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
ICMPv6 active timers:
  Last Router-Advertisement sent: 6 Secs
  Next Router-Advertisement sent in: 7 Secs
Router-Advertisement parameters:
  Periodic interval: 3 to 13 secs
  Router Preference: medium
  Send "Managed Address Configuration" flag: false
  Send "Other Stateful Configuration" flag: false
  Send "Current Hop Limit" field: 64
  Send "MTU" option value: 1500
  Send "Router Lifetime" field: 1900
  Send "Reachable Time" field: 0
  Send "Retrans Timer" field: 0
  Suppress RA: true
  Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
```

```
switch# show ipv6 nd interface

List of IPv6 Interfaces for VRF default
Interface vlan2 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent: 6 Secs
    Next Router-Advertisement sent in: 7 Secs
  Router-Advertisement parameters:
    Periodic interval: 3 to 13 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1900
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 nd interface prefix

show ipv6 nd interface prefix [all-vrfs | vrf <VRF-NAME>]

Description

Shows IPv6 prefix information for all VRFs or a specific VRF. If no options are specified, shows information for the default VRF.

Parameter	Description
all-vrfs	Shows prefix information for all VRFs.
vrf <VRF-NAME>	Name of a VRF.

Examples

Showing prefix information for the default VRF:

```
switch# show ipv6 nd interface prefix

List of IPv6 Interfaces for VRF default
List of IPv6 Prefix advertised on 1/1/1
  Prefix : 4545::/65
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

```
switch# show ipv6 nd interface prefix

List of IPv6 Interfaces for VRF default
List of IPv6 Prefix advertised on vlan2
  Prefix : 4545::/65
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

Showing information for VRF red:

```
switch# show ipv6 nd interface prefix vrf red

List of IPv6 Interfaces for VRF red
List of IPv6 Prefix advertised on 1/1/2
  Prefix : 2001::/64
  Enabled : Yes
  Validlife time : 2592000
```

```
Preferred lifetime : 604800
On-link : Yes
Autonomous : Yes
```

```
switch# show ipv6 nd interface prefix vrf red
```

```
List of IPv6 Interfaces for VRF red
List of IPv6 Prefix advertised on vlan3
  Prefix : 2001::/64
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 nd interface route

```
show ipv6 nd interface route [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays route information of all interfaces in the default VRF.

Parameter	Description
all-vrfs	Displays information about interfaces in all VRFs.
vrf <VRF-NAME>	Displays information about interfaces in a particular VRF. Or, if <VRF-NAME> is not specified, displays information for the default VRF.

Examples

Showing routing information for interface **1/1/1** in the default VRF:

```
switch# show ipv6 nd interface route
```

```
List of IPv6 Interfaces for VRF default
List of IPv6 Routes advertised on 1/1/1
  Route : 1::/64
  Enabled : Yes
```

```
Route lifetime : 200
Route preference : high
```

Showing routing information for interface **1/1/1** in VRF red:

```
switch# show ipv6 nd interface route vrf red

List of IPv6 Interfaces for VRF red
List of IPv6 Routes advertised on 1/1/2
  Route : 2::/64
  Enabled : No
  Route lifetime : 1800
  Route preference : low
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 nd ra dns search-list

```
show ipv6 nd ra dns search-list
```

Description

Displays domain name information on all interfaces.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com
switch# show ipv6 nd ra dns search-list
Recursive DNS Search List on: 1
  Suppress DNS Search List: Yes
  DNS Search 1: test.com    lifetime 1800
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 nd ra dns server

show ipv6 nd ra dns server

Description

Displays DNS server information on all interfaces.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: 1
    Suppress DNS Server List: Yes
    DNS Server 1: 2001::1      lifetime 1800
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

sFlow is a technology for monitoring traffic in switched or routed networks. The sFlow monitoring system is comprised of:

- An sFlow Agent that runs on a network device, such as a switch. The agent uses sampling techniques to capture information about the data traffic flowing through the device and forwards this information to an sFlow collector.
- An sFlow Collector that receives monitoring information from sFlow agents. The collector stores this information so that a network administrator can analyze it to understand network data flow patterns.

The sFlow UDP datagrams sent to a collector are not encrypted, therefore any sensitive information contained in an sFlow sample is exposed.

sFlow agent

The sFlow agent on the switch provides ingress sampling of all forwarded layer 2 and layer 3 traffic on LAG and Ethernet ports. High-availability is supported (packet sampling continues to work after switch-over).

The sFlow agent can communicate with up to three sFlow collectors at the same time. The agent communicates with collectors on the default VRF and non-default VRF.

Although you can configure very high sampling rates, the switch may drop samples if it cannot handle the rate of sampled packets. High sampling rates may also cause high CPU usage resulting in control plane performance issues.

A single sFlow datagram sent to a collector contains multiple flow and counter samples. The total number of samples an sFlow datagram can contain varies depending on the settings for header size and maximum datagram size.

Default settings

- sFlow is disabled on all interfaces.
- Collector port: UDP port 6343.
- sflow sampling mode: Ingress
- Sampling rate: 4096.
- Polling interval: 30 seconds.
- Header size: 128 bytes.
- Max datagram size: 1400 bytes.

Supported features

- Global sampling rate
- Interface counters polling

- Agent IP configuration for IPv4 and IPv6
- Header size configuration
- Max datagram size configuration
- Ingress sampling for all forwarded traffic (L2, L3)
- Enable/Disable sFlow per interface
- Support for three remote collectors
- An out-of-band collector can be defined on the management VRF
- A collector can be defined on the default and non-default VRF
- Sampling on Ethernet and LAG interfaces
- High availability support (sampling continues to work after switch-over)
- Source IP support (setting source IP for sFlow datagrams sent to a remote collector)

For the 4100i, 6000, and 6100 switch series, collector can be defined only on the default VRF. Also, there is no sampling of egress traffic.

Limitations

- Sampling rate cannot be set per interface (global only)
- sFlow is not configurable through SNMP.

Configuring the sFlow agent

Procedure

1. Configure one or more sFlow collectors with the command `sflow collector`. This determines where the sFlow agent sends sFlow information.
2. Enable the sFlow agent on all interfaces, or on a specific interface, with the command `sflow`.
3. Define the address of the sFlow agent with the command `sflow agent-ip`.
4. By default, the source IP address for sFlow datagrams is set to the IP address of the outgoing switch interface on which the sFlow client is communicating with a collector. Since the switch can have multiple routing interfaces, datagrams can potentially be sent on different paths at different times, resulting in different source IP addresses for the same client. To resolve this issue, define a single source IP address. For details, see *Single source IP address* in the *Fundamentals Guide*.
5. For most deployments, the default values for the following settings do not need to be changed. If your deployment requires different settings, change the default values with the indicated commands:

sFlow setting	Default value	Command to change it
Rate at which packets are sampled.	1 in every 4096 packets	<code>sflow sampling</code>
Rate at which the switch sends data to an sFlow collector.	30 seconds	<code>sflow polling</code>
Size of the sFlow header.	128 bytes	<code>sflow header-size</code>

sFlow setting	Default value	Command to change it
Maximum size of an sFlow datagram.	1400 bytes	sflow max-datagram-size

- Review sFlow configuration settings with the command `show sflow`.

Example

This example creates the following configuration:

- Configures an sFlow collector with the IP address **10.10.20.209**.
- Enables the sFlow agent on all interfaces.
- Defines the sFlow agent IP address to be **10.10.1.5**.

```
switch(config)# sflow collector 10.10.20.209
switch(config)# sflow
switch(config)# sflow agent-ip 10.0.0.1
```

sFlow scenario

In this scenario, two hosts send sFlow traffic through a switch to an sFlow collector.

Procedure

- Enable sFlow globally.

```
switch# config
switch(config)# sflow
```
- Set the sFlow agent IP address to **10.10.12.1**.

```
switch(config)# sflow agent-ip 10.10.12.1
```
- Set the sFlow collector IP address to **10.10.12.2**.

```
switch(config)# sflow collector 18.2.2.2
```
- Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 20
```
- Configure interface **vlan 2** with IP address **10.10.10.1/24**.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.10.1/24
switch(config)# quit
```
- Configure interface **vlan 3** with IP address **10.10.11.1/24**.

```
switch(config)# interface vlan 3
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.11.1/24
switch(config)# quit
```
- Configure interface **vlan 4** with IP address **10.10.12.1/24**.

```
switch(config)# interface vlan 4
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config)# quit
```
- Verify sFlow configuration

```

switch# show sflow

sFlow Global Configuration
-----

sFlow                               enabled
Collector IP/Port/Vrf               10.10.10.2/6343/default
Agent Address                       10.0.0.1
Sampling Rate                       1024
Polling Interval                    30
Header Size                         128
Max Datagram Size                   1400


sFlow Status
-----

Running - Yes


sFlow enabled on Interfaces:
-----

lag100


sFlow Statistics
-----

Number of Samples                   200

```

sFlow scenario 2

In this scenario, two hosts connected to different switches send sFlow traffic to a collector. A LAG is used to connect the two switches.

Procedure

1. Configure switch 1.
 - a. Enable sFlow globally.

```
switch# config
switch(config)# sflow
```
 - b. Set the sFlow agent IP address to **10.10.12.1**.

```
switch(config)# sflow agent-ip 10.10.12.1
```
 - c. Set the sFlow collector IP address to **10.10.12.2**.

```
switch(config)# sflow collector 10.10.12.2
```

- d. Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 10
```
- e. Create VLAN 8.

```
switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit
```
- f. Define LAG 100 and assign VLAN **vlan 8** to it.

```
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 8
switch(config-lag-if)# lacp mode active
```
- g. Configure interface **vlan 5**.

```
switch(config)# interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-lag-if)# no routing
switch(config-if-vlan)# vlan access 8
```
- h. Configure VLANs **4** and **5** as members of LAG **100**.

```
switch# (config)#interface vlan 4
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# lag 100
switch(config-if-vlan)# exit
switch(config)# interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# lag 100
switch(config-if-vlan)# exit
```
- i. Configure interface **vlan 5** with IP address **10.10.12.1/24**.

```
switch# (config)#interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config-if-vlan)# quit
```
- j. Verify sFlow configuration.

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                enabled
Collector IP/Port/Vrf 10.10.10.2/6343/default
Agent Address         10.0.0.1
Sampling Rate         1024
Polling Interval      30
Header Size           128
Max Datagram Size     1400

sFlow Status
-----
Running - Yes

sFlow enabled on Interfaces:
-----
lag100

sFlow Statistics
-----
Number of Samples      200
```

2. Configure switch 2.

a. Create VLAN 8.

```
switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit
```

b. Define LAG 100 and assign VLAN **vlan 8** to it.

```
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 8
switch(config-lag-if)# lacp mode active
```

c. Configure interface 1/1/1.

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 8
```

d. Configure interface 1/1/2 and 1/1/3 as members of LAG 100.

```
switch# (config)#interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
switch(config)-if# interface 1/1/3
switch(config-if)# no shutdown
switch(config-if)# lag 100
switch(config-if)# exit
```

sFlow agent commands

clear sflow statistics

```
clear sflow statistics {global | interface <INTERFACE-NAME>}
```

Description

This command clears the sFlow sample statistics counter to 0 either globally or for a specific interface.

Parameter	Description
global	Specifies all interfaces on the switch.
interface <INTERFACE-NAME>	Specifies the name of an interface on the switch.

Examples

Clearing the global sFlow sample statistics counter to 0 globally:

```
switch(config)# clear sflow statistics global
```

Clearing the global sFlow sample statistics counter to 0 for interface 1/1/1:

```
switch(config)# clear sflow statistics interface 1/1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

sflow

sflow
no sflow

Description

Enables the sFlow agent.

- In the **config** context, this command enables the sFlow agent globally on all interfaces.
- In an **config-if** context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The **no** form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

Examples

Enabling sFlow globally on all interfaces:

```
switch(config)# sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config)# no sflow
```

Enabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# sflow
```

Disabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# no sflow
```

Enabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# sflow
```

Disabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config config-if	Administrators or local user group members with execution rights for this command.

sflow agent-ip

```
sflow agent-ip <IP-ADDR>
no sflow agent-ip [<IP-ADDR>]
```

Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The **no** form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. The agent address is used to identify the switch in all sFlow datagrams sent to sFlow collectors. It is usually set to an IP address on the switch that is reachable from an sFlow collector.

Examples

Setting the agent address to **10.10.10.100**:

```
switch(config)# sflow agent-ip 10.0.0.100
```


Setting the agent address to **2001:0db8:85a3:0000:0000:8a2e:0370:7334**:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow collector

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]  
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

Parameter	Description
collector <IP-ADDR>	Specifies the IP address of a collector in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
port <PORT>	Specifies the UDP port on which to send information to the sFlow collector. Range: 0 to 65536. Default: 6343.
vrf <VRF>	Specifies the VRF on which to send information to the sFlow collector. The VRF must be defined on the switch. If no VRF is specified, the default VRF (default) is used.

Example

Defining a collector with IP address **10.10.10.100** on UDP port **6400**:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow disable

sflow disable

Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow header-size

sflow header-size <SIZE>
no sflow header-size [<SIZE>]

Description

Sets the sFlow header size in bytes.

The **no** form of this command sets the header size to the default value of 128.

Parameter	Description
header-size <SIZE>	Specifies the sFlow header size in bytes. Range: 64 to 256. Default: 128.

Examples

Setting the header size to **64** bytes:

```
switch(config)# sflow header-size 64
```

Setting the header size to the default value of **128** bytes:

```
switch(config)# no sflow header-size
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow max-datagram-size

```
sflow max-datagram-size <SIZE>  
no sflow max-datagram-size [<SIZE>]
```

Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The **no** form of this command sets maximum number of bytes to the default value of 1400.

Parameter	Description
max-datagram-size <SIZE>	Specifies the maximum datagram size in bytes. Range: 1 to 9000. Default: 1400.

Examples

Setting the datagram size to **1000** bytes:

```
switch(config)# sflow max-datagram-size 1000
```

Setting the header size to the default value of **1400** bytes:

```
switch(config)# no sflow max-datagram-size
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow polling

```
sflow polling <INTERVAL>
no sflow polling [<INTERVAL>]
```

Description

Defines the global polling interval for sFlow in seconds.

The **no** form of this command sets the polling interval to the default value of 30 seconds.

Parameter	Description
<INTERVAL>	Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30.

Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

sflow sampling

```
sflow sampling <RATE>
```

```
no sflow sampling [<RATE>]
```

Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The **no** form of this command sets the sampling rate to the default value of 4096.

Parameter	Description
<code>sampling <RATE></code>	Specifies the sampling rate. Range: 1 to 16777215. Default: 4096.

Examples

Setting the sampling rate to **5000**:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to **1000**:

```
switch(config)# sflow sampling 1000
Setting the sFlow sampling rate lower than 4096 is not recommended and might
affect system performance.
Do you want to continue [y/n]? y
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show sflow

```
show sflow [interface <INTERFACE-NAME>]
```

Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface. It also displays the current status of sFlow on the device and reports any errors that require attention.

If sFlow is enabled on the interfaces associated with a lag interface, then the interfaces will not be shown as separate entries under sFlow enabled on Interface in the output. Only the associated lag interface will have an entry in the column.

Parameter	Description
interface <INTERFACE-NAME>	Specifies the name of an interface on the switch.

Examples

Showing sFlow information for all interfaces:

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                      enabled
Collector IP/Port/Vrf      10.10.10.2/6343/default
Agent Address              10.0.0.1
Sampling Rate              1024
Polling Interval           30
Header Size                128
Max Datagram Size          1400

sFlow Status
-----
Running - Yes

sFlow enabled on Interfaces:
-----
lag100

sFlow Statistics
-----
Number of Samples          200
```

Showing sFlow information for interface **1/1/1**:

```
switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1
-----
sFlow                      enabled
Sampling Rate              1024
Number of Samples          30
sFlow Sampling Status      success
```

Showing sFlow information for interface **lag 10**:

```
switch# show sflow interface lag 10
sFlow Configuration - Interface lag10
-----
sFlow                      enabled
Sampling Rate              4096
Number of Samples          0
sFlow Sampling Status      error
```

```
Sampling Status on LAG members
```

```
-----  
Intf 1/1/2          no agent  
Intf 1/1/3          no agent
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

The Dynamic Host Configuration Protocol (DHCP) enables the automatic assignment of IP addresses and other configuration settings to network devices.

On 4100i, 6000 and 6100 series switches, only SVIs are supported (ROP is not supported on these platforms). Hence, you need to first configure the dependent DHCP feature on a vlan, and then configure the vlan as an access vlan for a physical interface.

Example to configure the vlan as an access vlan for a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config-if)# interface vlan 100
switch(config-if-vlan)# <DHCP feature CLI>
```

DHCP client

By default, the switch operates as a DHCP client on VLAN 1 allowing it to automatically obtain an IP address from a DHCP server on the network to which it is connected.

Protocol and feature details

DHCP Client is supported on IPv4.

The IP DHCP assignment with HPE Aruba Networking AOS-CX switch is to allow the following:

- IP switch assignment.
- Managing the initial IP access to the switch.
- Prepare for Zero Touch Provisioning (ZTP) with HPE Aruba Networking Central.

Zero Touch Provisioning (ZTP) is only supported for IPV4 in HPE Aruba Networking Central

- Prepare for Zero Touch Provisioning (ZTP) by using DHCP options along with a TFTP server.

Supported platform and standards

The following table list the supported VLANs and ports for the AOS-CX Switch.

Table 1: *Supported VLAN and Ports for DHCP*

Platform	VLAN 1	Any Other VLAN	In band Port	Management VRF port
4100i	Yes	Yes	Yes	N/A

Platform	VLAN 1	Any Other VLAN	In band Port	Management VRF port
5400	Yes	Yes	Yes	Yes
6000	Yes	Yes	Yes	N/A
6100	Yes	Yes	Yes	N/A
6200	Yes	Yes	Yes	Yes
6300 6300L	Yes	Yes	Yes	Yes
6400	Yes	Yes	Yes	Yes
8320	No	No	No	Yes
8325 8325H 8325P	No	No	No	Yes
8360	No	No	No	Yes
8400	No	No	No	Yes
9300 9300S	No	No	No	Yes
10000	No	No	No	Yes

Configuration task list

To run the DHCP Client do the following:

- Enter the interface VLAN or Management VRF context
- Enable DHCP

Considerations and best practices

The IP DHCP can only be configured on one VLAN per switch.

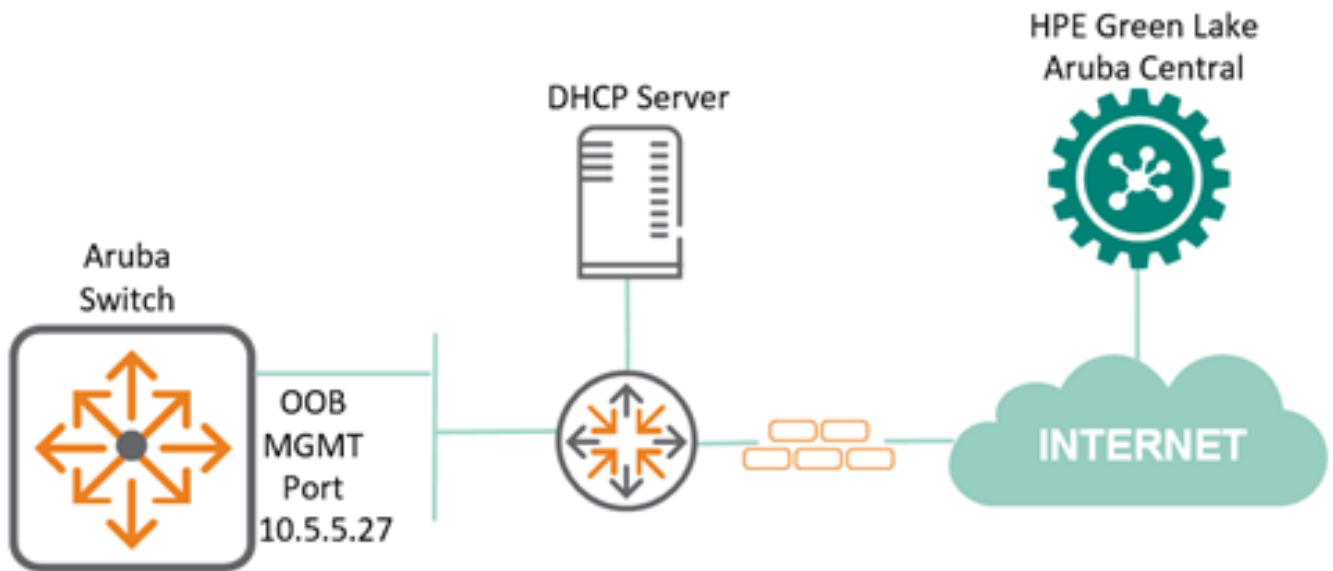
If an OOB Management port is available, you can configure the DHCP service simultaneously for the VLAN IN Band and the OOB Management port. When both the IN Band data port and the OOB Management port receive a DHCP address, then the DHCP address received on the OOBM port is preferred over the IN Band data port for ZTP.

Use case

IP DHCP assignment with HPE Aruba Networking AOS-CX switches is to allow IP switch assignment and Zero Touch Provisioning using HPE Aruba Networking Central, as shown below.

In the following DHCP client scenario the OOB Management port is in use. The IN Band data ports can also be used for Zero Touch Provisioning (ZTP).

For more information related to the supported ports for the different switches, see [Table 1, Supported VLAN and Ports for DHCP](#).



1. Switch configuration from factory on OOB Management port.

```
switch#  
!  
interface mgmt  
    no shutdown  
    ip dhcp  
!
```

2. OOB Management port gets assigned with an IPv4 address 10.5.5.27 from DHCP server.

```
switch# show interface mgmt  
Address Mode           : dhcp  
Admin State            : up  
Link State              : up  
Mac Address             : 88:3a:30:9a:39:01  
IPv4 address/subnet-mask : 10.5.5.27/24  
Default gateway IPv4    : 10.5.5.254  
IPv6 address/prefix     : 2a00:23c5:ac8a:3e01:8a3a:30ff:fe9a:3901/64  
IPv6 link local address/prefix: fe80::8a3a:30ff:fe9a:3901/64  
Default gateway IPv6    : fe80::f286:20ff:fe0b:fe5  
Primary Nameserver      : 10.5.19.69  
Secondary Nameserver    : 10.5.19.79  
Tertiary Nameserver     : 8.8.4.4
```

The following output shows a switch connected to HPE Aruba Networking Central allocated with an IP address of 10.5.5.27 on its OOB MGMT port by a DHCP server. Based on its designated configuration in the public cloud, the switch can get its configuration parameters directly from HPE Aruba Networking Central.

```

switch# show hpe_anw-central
  Central admin state           : enabled
  Central location              : device-prod2-
d2.central.arubanetworks.com
  VRF for connection           : mgmt
  Central connection status     : connected
  Central source                : activate
  Central source connection status : connected
  Central source last connected on : Wed Sep 22 18:39:05 UTC 2021
  System time synchronized from Activate : True
  Activate Server URL           : devices-v2.arubanetworks.com
  CLI location                  : N/A
  CLI VRF                       : N/A
  Source IP                     : 10.5.5.27
  Source IP Overridden          : False
  Central support mode          : disabled

```

DHCP client commands

ip dhcp

```

ip dhcp
no ip dhcp

```

Description

Enables the DHCP client on any interface VLAN to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled on VLAN 1.

The **no** form of the command disables DHCP mode and is supported only on interface VLANs.

Examples

Enabling the DHCP client on the interface vlan 1:

```

switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no shutdown

```

Disabling the DHCP client on the interface vlan 1:

```

switch(config)# interface vlan 1
switch(config-if-vlan)# no ip dhcp

```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

`ip dhcp` is supported only on one vlan at a time.

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 dhcp

ipv6 dhcp
no ipv6 dhcp

Description

Configure to enable DHCPv6 mode on the preferred VLAN. In DHCPv6 Mode, the VLAN will get DHCP-Server allocated IPv6 address along with DNS and NTP attributes.

The **no** form of the command disables DHCP mode and is supported only on interface VLANs.

ipv6 dhcp is supported only on one vlan at a time. For dhcpv6 client to get an ipv6 address, ipv6 linklocal address should be configured.

ipv6 dhcp is ignored when static ipv6 address is configured.

Examples

Enabling the DHCPv6 client on the interface vlan 4 under non-default VRF:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# vrf attach red
switch(config-if-vlan)# ipv6 dhcp
```

Disabling the DHCPv6 client on the interface vlan 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 dhcp
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.15	Command was introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip dhcp option

```
ip dhcp option [host-name | broadcast-flag]
no ip dhcp option [host-name | broadcast-flag]
```

Description

This command enables the DHCP client host name and broadcast flag globally.

If the **ip dhcp option broadcast-flag** command is enabled, then the DHCP offer and ack packets in the DHCP requests will be treated as broadcast packets. These packets will not be forwarded due to the presence of a default static route.

The **no** form of this command globally disables the host name and DHCP client broadcast flag options.

The **ip dhcp option broadcast-flag** command should be configured before configuring the **ip dhcp** command.

Example

Enabling the DHCP client broadcast flag globally:

```
switch(config)# ip dhcp option
broadcast-flag DHCP Client broadcast-flag option (Default:disabled)
host-name DHCP Client hostname option
switch(config)# ip dhcp option
% Command incomplete.
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# ip dhcp op
% There is no matched command.
switch(config-if-vlan)# ip dhcp op
Invalid input: ip
switch(config-if-vlan)#
```

Enabling the DHCP client host name globally:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp option host-name
```

Disabling the DHCP client broadcast flag globally:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip dhcp option broadcast-flag
```

Disabling the DHCP client host name globally:

```
switch(config)# ip dhcp option
broadcast-flag DHCP Client broadcast-flag option (Default:disabled)
host-name DHCP Client hostname option
switch(config)# ip dhcp option
% Command incomplete.
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no ip dhcp option host-name
% There is no matched command.
switch(config-if-vlan)# ip dhcp op
```

```
Invalid input: ip
switch(config-if-vlan)#
```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.13.1000	Command Introduced

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

show ip dhcp

```
show ip dhcp
```

Description

Displays DHCP IPv4 information on the ports.

Examples

Displaying the DHCP IPv4 information on the ports:

```
switch# show ip dhcp
DHCP Options: Broadcast-flag, Hostname

INTERFACE-NAME  ADDRESS          DEFAULT_GATEWAY  DOMAIN_NAME  VRF  DNS-SERVERS
-----
vlan1           10.254.239.10/27          domain.com      default      50.0.0.2,
50.0.0.3, 50.0.0.4
```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.13.1000	The output parameters, Broadcast-flag and Hostname were introduced.
10.09 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 dhcp

show ipv6 dhcp

Description

This command displays DHCP IPv6 information on the ports.

Examples

Displaying the DHCP IPv6 information on the ports:

```
6410(config)# show ipv6 dhcp

VRF default
-----
Interface vlan1
-----
IPv6 address      : 3013::1050/64
Domain search     : paramv6.org, test1.in, test2.in, test3.in, test4.in, test5.in
DNS servers       : 3013::100, 3013::101, 3013::102
NTP servers       : 3013::201, 3013::202
```

Command History

Release	Modification
10.16.1000	Support added for the 8100, 832x, 8360, 8400, 9300, 9300S, and 10000 Switch Series.
10.15	Command was introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

DHCP relay agent

The function of the DHCP relay agent is to forward the DHCP messages to other subnets so that the DHCP server does not have to be on the same subnet as the DHCP clients. The DHCP relay agent transfers DHCP messages from the DHCP clients located on a subnet without a DHCP server, to other subnets. It also relays answers from DHCP servers to DHCP clients.

Supported platform and standards

The following table list the supported platforms for DHCPv4 relay and DHCPv6 relay.

Table 1: *Support platforms for DHCPv4 relay and DHCPv6 relay*

Platform	DHCPv4 Relay	DHCPv4 Smart Relay	DHCPv6 Relay
4100i	Yes	No	Yes
6000	Yes	No	Yes
6100	Yes	No	Yes

Table 2: *Scale*

Platform	DHCP v4/v6 enabled SVI	DHCP v4/v6 helper SVI
4100i	16	8
6000	16	8
6100	16	8

DHCP relay behaviors are as follows:

- DHCPv6 relay is disabled by default.
- DHCP relay hop count is enabled by default.
- In DHCPv4 relay the option 82 policy is replace by default.
- The MAC address of the switch is used as the option 82 remote ID by default.
- In DHCPv6 relay the option 79 policy is disabled by default.
- In DHCPv4 relay the source-interface is disabled by default.

Protocol and feature details

Supported interfaces

The DHCP relay agent is supported on layer 3 VLAN interfaces, and LAG interfaces. DHCP relay is not supported on the management interface.

DHCPv4 relay agent

Hop count in DHCP requests

When a DHCP client broadcasts request, the DHCP relay agent in the switch receives the packets and forwards them to the DHCP server as unicast requests. During this process, the DHCP relay agent increments the hop count before forwarding DHCP packets to the server. The DHCP server, in turn, includes the hop count in the DHCP header in the response sent back to a DHCP client.

DHCP relay option 82

Option 82 is called the relay agent information option. When a DHCP relay agent forwards client-originated DHCP packets to a DHCP server, the option 82 field is inserted/replaced, or the packet with this option is dropped. Servers recognizing the relay agent information option may use the information to implement policies for the assignment of IP addresses and other parameters. The relay agent relays the server-to-client replies to the client.

If a second relay agent is configured to add its own option 82 information, it can encapsulate option 82 information in messages from a first relay agent. The DHCP server uses the option 82 information from both relay agents to decide the IP address for the client.

A DHCP relay option 82 source-interface needs to be enabled when a source interface is used for an Intra-VRF deployment. In this deployment type, the client and DHCP server are in same VRF, but the relay will use the source interface.

Configuring a BOOTP/DHCP relay gateway

The DHCP relay agent selects the lowest-numbered IP address on the interface to use for DHCP messages. The DHCP server then uses this IP address when it assigns client addresses. However, this IP address may not be the same subnet as the one on which the client needs the DHCP service. This feature provides a way to configure a gateway address for the DHCP relay agent to use for relayed DHCP requests, rather than the DHCP relay agent automatically assigning the lowest-numbered IP address.

Configuring the DHCPv4 relay agent

Prerequisites

- An enabled layer 3 VLAN

Procedure

1. The DHCPv4 relay agent is enabled by default. If it was previously disabled, enable it with the command `dhcp-relay`.
2. Configure one or more IP helper addresses with the command `ip helper-address`. This determines where the DHCPv4 relay agent forwards DHCP requests. IP helper addresses can be configured on layer 3 interfaces, layer 3 VLAN interfaces, and LAG interfaces.
3. If you want to modify the content of forwarded DHCP packets or drop DHCP packets, configure option 82 support with the command `dhcp-relay option 82`.
4. Define the gateway address that the DHCPv4 relay agent will use with the command [ip bootp-gateway](#).
5. If required, enable the hop count increment feature with the command `dhcp-relay hop-count-increment`.
6. Review DHCPv4 relay agent configuration settings with the commands `show dhcp-relay`, `show ip helper-address`, and `show dhcp-relay bootp-gateway`.

Example

This example creates the following configuration:

- Enables the DHCPv4 relay agent.
- Enables interface **1/1/1** and assigns an IPv4 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **10.10.20.209** on the interface.
- Enables DHCP option 82 support and replaces all option 82 information with the values from the switch with the switch MAC address as the remote ID.

On 4100i, 6000 and 6100 series switches, only SVIs are supported.

```

switch(config)# dhcp-relay
switch(config)# interface 1/1/1
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 198.51.100.1/24
switch(config-if-vlan)# ip helper-address 10.10.20.209
switch(config-if-vlan)# exit
switch(config)# dhcp-relay option 82 replace mac
switch# show dhcp-relay

```

```

DHCP Relay Agent           : enabled
DHCP Request Hop Count Increment : enabled
Option 82                  : disabled
Source-Interface           : disabled
Response Validation        : disabled
Option 82 Handle Policy    : replace
Remote ID                   : mac

```

DHCP Relay Statistics:

Valid Requests	Dropped Requests	Valid Responses	Dropped Responses
60	10	60	10

DHCP Relay Option 82 Statistics:

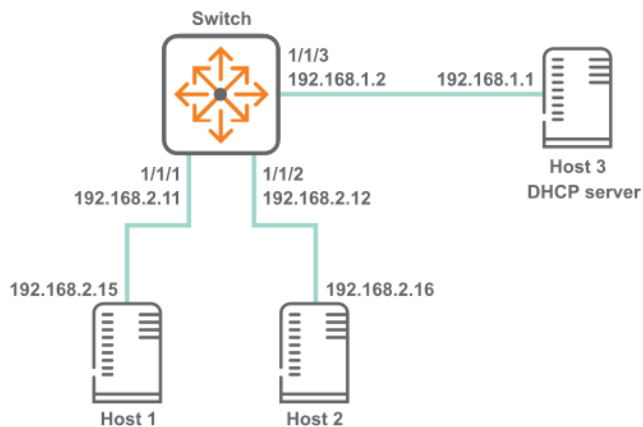
Valid Requests	Dropped Requests	Valid Responses	Dropped Responses
50	8	50	8

Use Case

This section explain the use cases for DHCPv4 relay agent.

DHCPv4 relay scenario 1

In this scenario, DHCP relay on the switch enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



Procedure

1. DHCP relay is enabled by default. If it was previously disabled, enable it.

```
switch# config  
switch(config)# dhcp-relay
```

2. Define an IPv4 helper address on interface vlan 100

```
switch(config)# interface 1/1/1  
switch(config-if)# vlan access 100  
switch(config-if)# exit  
switch(config)# interface 1/1/2  
switch(config-if)# vlan access 100  
switch(config-if)# exit  
switch(config)# interface vlan 100  
switch(config-if-vlan)# ip address 192.168.2.11/24  
switch(config-if-vlan)# ip helper-address 192.168.1.1
```

3. Verify DHCP relay configuration.

```
switch# show dhcp-relay
```

DHCP Relay Agent	:	enabled
DHCP Request Hop Count Increment	:	enabled
Option 82	:	disabled
Source-Interface	:	disabled
Response Validation	:	disabled
Option 82 Handle Policy	:	replace
Remote ID	:	mac

DHCP Relay Statistics:

Valid Requests	Dropped Requests	Valid Responses	Dropped Responses
60	10	60	10

DHCP Relay Option 82 Statistics:

Valid Requests	Dropped Requests	Valid Responses	Dropped Responses
50	8	50	8

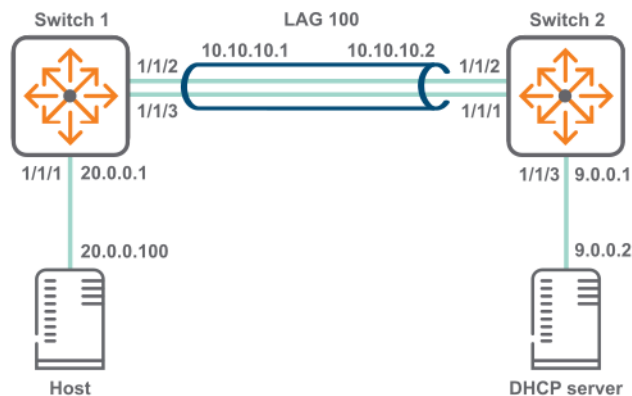
```
switch(config)# show ip helper-address
```

IP Helper Addresses

Interface: vlan100	
IP Helper Address	VRF
192.168.1.1	default

DHCPv4 relay scenario 2

In this scenario, host on switch 1 reaches the DHCP server on switch two via a LAG. The physical topology of the network looks like this:



Procedure

1. On switch 1:
 - a. Create LAG 100 and assign an IP address to it.

```
switch# config
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 200
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# exit
switch(config)# interface vlan 200
switch(config-if-vlan)# ip address 10.0.10.1/24
```

- b. Assign an IP address to interface vlan 100 and an IP helper address to reach the DHCP server.

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 100
switch(config-if)# exit
switch(config)# interface vlan 100
switch(config-if-vlan)# ip address 20.0.0.1/8
switch(config-if-vlan)# ip helper-address 9.0.0.2
```

- c. Assign interfaces 1/1/2 and 1/1/3 to LAG 100.

```
switch(config-if)# interface 1/1/2
switch(config-if)# lag 100
switch(config-if)# interface 1/1/3
switch(config-if)# lag 100
```

- d. Create a route between 10.0.10.2 and 9.0.0.0.

```
switch(config)# ip route 9.0.0.0/24 10.0.10.2
```

2. On switch 2:
 - a. Create LAG 100 and assign an IP address to it.

```
switch# config
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 200
switch(config-lag-if)# lACP mode active
switch(config-lag-if)# exit
switch(config)# interface vlan 200
switch(config-if-vlan)# ip address 10.0.10.2/24
```

- b. Assign an IP address to interface vlan 300.

```
switch(config)# interface 1/1/3
switch(config-if)# no shutdown
switch(config-if)# vlan access 300
switch(config-if)# exit
switch(config)# interface vlan 300
switch(config-if-vlan)# ip address 9.0.0.1/24
```

- c. Assign interfaces 1/1/1 and 1/1/2 to LAG 100.

```
switch(config-if)# interface 1/1/1
switch(config-if)# lag 100
switch(config-if)# interface 1/1/2
switch(config-if)# lag 100
```

- d. Create a route between 20.0.0.0 and 10.0.10.1.

```
switch(config)# ip route 20.0.0.0/8 10.0.10.1
```

DHCPv4 relay commands

dhcp-relay

```
dhcp-relay
no dhcp-relay
```

Description

Enables DHCP relay support. DHCP relay is enabled by default. DHCP relay is not supported on the management interface.

The **no** form of this command disables DHCP relay (and DHCP relay option 82) support.

Examples

This example enables DHCP relay support.

```
switch(config)# dhcp-relay
```

This example removes DHCP relay support.

```
switch(config)# no dhcp-relay
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-relay hop-count-increment

```
dhcp-relay hop-count-increment
no dhcp-relay hop-count-increment
```

Description

Enables the DHCP relay hop count increment feature, which causes the DHCP relay agent to increment the hop count in all relayed DHCP packets. Hop count is enabled by default.

The **no** form of this command disables the hop count increment feature.

Examples

Enabling the hop count increment feature.

```
switch(config)# dhcp-relay hop-count-increment
```

Disabling the hop count increment feature.

```
switch(config)# no dhcp-relay hop-count-increment
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-relay option 82

```
dhcp-relay option 82 {replace [validate] | drop [validate] |
    keep | source-interface | validate [replace | drop]} [ip | mac]
no dhcp-relay option 82 {replace [validate] | drop [validate] |
    keep | source-interface | validate [replace | drop]} [ip | mac]
```

Description

Configures the behavior of DHCP relay option 82. A DHCP relay agent can receive a message from another DHCP relay agent having option 82. The relay information from the previous relay agent is replaced by default.

The **no** form of this command disables the DHCP relay option 82 configurations. Option 82 is disabled when DHCP relay is disabled globally. When DHCP relay is re-enabled, option 82 also needs to be re-enabled using the **dhcp-relay option 82** command.

DHCP Relay is supported over VXLAN with both IPv4 and IPv6 underlay.

Parameter	Description
replace	Replace the existing option 82 field in an inbound client DHCP packet with the information from the switch. The remote ID and circuit ID information from the first relay agent is lost. Default.
validate	Validate option 82 information in DHCP server responses and drop invalid responses.
drop	Drop any inbound client DHCP packet that contains option 82 information.
keep	Keep the existing option 82 field in an inbound client DHCP packet. The remote ID and circuit ID information from the first relay agent is preserved.
source-interface	Configures the DHCP relay to use a configured source IP address for inter-VRF server reachability. Set the source IP address with the command <code>ip source-interface</code> .
ip	Use the IP address of the interface on which the client DHCP packet entered the switch as the option 82 remote ID.
mac	Use the MAC address of the switch as the option 82 remote ID. Default.

Example

This example enables DHCP option 82 support and replaces all option 82 information with the values from the switch, with the switch MAC address as the remote ID.

```
switch(config)# dhcp-relay option 82 replace mac
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution

Platforms	Command context	Authority
6000 6100		rights for this command.

diag-dump dhcp-relay basic

diag-dump dhcp-relay basic

Description

Dumps DHCP relay configurations for all interfaces.

Examples

This example enables DHCP relay support.

```
switch# diag-dump dhcp-relay basic
=====
[Start] Feature dhcp-relay Time : Sun Apr 26 06:38:10 2020
=====
-----
[Start] Daemon hpe-relay
-----
DHCP Relay : 1
DHCP Relay hop-count-increment : 1
DHCP Relay Option82 : 1
DHCP Relay Option82 validate : 0
DHCP Relay Option82 policy : keep
DHCP Relay Option82 remote-id : mac
DHCP Relay Option82 Source Intf : Disable
System Mac [f4:03:43:80:27:00]
VRF :BLUE, Source Ip:200.0.0.10
vsx: Not Present
Interface vlan2: 1
Client Packet Statistics:
Valid Dropped O82_Valid O82_Dropped vsx_drops
-----
0 0 0 0 0
Server Packet Statistics:
Valid Dropped O82_Valid O82_Dropped Invalid_IP_Drops To_Dsnoop
-----
0 0 0 0 0 0
client request dropped packets with extn option 82 = 0
client request valid packets with extn option 82 = 0
server request dropped packets with extn option 82 = 0
server request valid packets with extn option 82 = 0
Port 67 - 200.0.0.100,2
source vrf-BLUE.
Interface vlan3: 1
Client Packet Statistics:
Valid Dropped O82_Valid O82_Dropped vsx_drops
-----
0 0 0 0 0
Server Packet Statistics:
Valid Dropped O82_Valid O82_Dropped Invalid_IP_Drops To_Dsnoop
-----
0 0 0 0 0 0
client request dropped packets with extn option 82 = 0
client request valid packets with extn option 82 = 0
server request dropped packets with extn option 82 = 0
server request valid packets with extn option 82 = 0
```



```

Port 67 - 200.0.0.100,2
source vrf-BLUE.
[End] Daemon hpe-relay
=====
[End] Feature dhcp-relay
=====
Diagnostic-dump captured for feature dhcp-relay

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ip bootp-gateway

```

ip bootp-gateway <IPV4-ADDR>
no ip bootp-gateway <IPV4-ADDR>

```

Description

Configures a gateway address for the DHCP relay agent to use for DHCP requests. By default DHCP relay agent picks the lowest-numbered IP address on the interface.

The **no** form of this command removes the gateway address.

Parameter	Description
<IPV4-ADDR>	Specifies the IP address of the gateway in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

Examples

Setting the IP address of the gateway for interface vlan 100 to **10.10.10.10**:

```

switch(config)# interface vlan100
switch(config-if)# ip bootp-gateway 10.10.10.10

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

ip helper-address

```
ip helper-address <IPv4-ADDR> [vrf <VRF-NAME>]
no ip helper-address <IPv4-ADDR> [vrf <VRF-NAME>]
```

Description

Defines the address of a remote DHCP server or DHCP relay agent. Up to eight addresses can be defined. The DHCP relay agent forwards DHCP client requests to all defined servers.

If IP helper address is defined with VRF argument then this command requires you define a source IP address for DHCP relay with the command `ip source-interface`. The configured source IP on the VRF is used to forward DHCP packets to the server.

A helper address cannot be defined on the OOBM interface.

The **no** form of this command removes an IP helper address.

Parameter	Description
helper-address <IPv4-ADDR>	Specifies the helper IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
vrf <VRF-NAME>	Specifies the name of a VRF. Default: default .

Examples

Defining the IP helper address 10.10.10.209 on interface vlan 100:

```
switch(config)# interface vlan100
switch(config-if-vlan)# ip helper-address 10.10.10.209
```

Removing the IP helper address 10.10.10.209 on interface vlan 100:

```
switch(config-if-vlan)# no ip helper-address 10.10.10.209
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

show dhcp-relay

show dhcp-relay

Description

Shows DHCP relay configuration settings.

Example

Showing DHCP relay settings:

```
switch(config)# show dhcp-relay

DHCP Relay Agent           : enabled
DHCP Smart Relay          : enabled
DHCP Request Hop Count Increment : enabled
Option 82                  : enabled
Source-Interface           : disabled
Response Validation        : disabled
Option 82 Handle Policy    : replace
Remote ID                  : mac

DHCP Relay Statistics:

Valid Requests  Dropped Requests  Valid Responses  Dropped Responses
-----
60              10                 60              10

DHCP Relay Option 82 Statistics:

Valid Requests  Dropped Requests  Valid Responses  Dropped Responses
-----
50              8                 50              8
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	

show dhcp-relay bootp-gateway

show dhcp-relay bootp-gateway [interface <INTERFACE-NAME>]

Description

Shows the bootp gateway defined for all interfaces or a specific interface.

Parameter	Description
<INTERFACE-NAME>	Specifies an interface. Format: member/slot/port.

Examples

Showing the designated bootp gateway for all interfaces:

```
switch(config)# show dhcp-relay bootp-gateway

BOOTP Gateway Entries

Interface          Source IP
-----
vlan10             1.1.1.1
vlan20             2.2.2.2
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	

show ip helper-address

```
show ip helper-address [interface <INTERFACE-ID>]
```

Description

Shows the IP helper addresses defined for all interfaces or a specific interface.

Parameter	Description
interface <INTERFACE-ID>	Specifies an interface. Format: member/slot/port.

Examples

Showing the IP helper addresses for all interfaces:

```
switch(config)# show ip helper-address

IP Helper Addresses

Interface: vlan11
  IP Helper Address          VRF
-----

```

```

10.10.10.209                default
Interface: vlan20
IP Helper Address           VRF
-----
3.3.3.3                     default
20.20.20.20                 default

```

Showing the IP helper addresses for interface vlan20:

```

switch(config)# show ip helper-address interface vlan20

IP Helper Addresses

Interface: vlan20
IP Helper Address           VRF
-----
3.3.3.3                     default
20.20.20.20                 default

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	

DHCPv6 relay agent

DHCPv6 relay VRF support

DHCPv6 relay is VRF aware. DHCPv6 client requests received on an interface are forwarded to the configured servers via that interface's VRF. DHCPv6 server responses received on an interface are forwarded to the client which is reachable via that interface's VRF.

DHCPv6 relay does not have inter-VRF support.

Limitations

DHCPv6 Relay requires an egress interface to be configured for the IPv6 multicast helper address. DHCPv6-Relay multicast helper configuration is not allowed over VxLAN tunnels.

The following configuration is required in order to use source interface for DHCPv6 relay. Configuring the IPv6 source-interface interface to use for DHCPv6 relay:

```
switch(config)# ipv6 source-interface dhcp_relay interface loopback3 vrf test1
```

Enabling DHCPv6 relay to use the configured source interface:

```
switch(config)# dhcpv6-relay source-interface
```

Configuring the DHCPv6 relay agent

Prerequisites

- An enabled layer 3 VLAN

Procedure

1. Enable the DHCPv6 relay agent with the command `dhcpv6-relay`.
2. Configure one or more IP helper addresses with the command `ipv6 helper-address`. This determines where the DHCPv6 agent forward DHCP requests.
3. If you want to enable DHCP option 79 support to forward client link-layer addresses, use the command `dhcpv6-relay option 79`.
4. Review DHCPv6 relay agent configuration settings with the commands `show dhcpv6-relay` and `show ipv6 helper-address`.

Example

This example creates the following configuration:

- Enables the DHCPv6 relay agent.
- Enables interface **1/1/2** and assigns an IPv6 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **FF01::1:1000** on interface **1/1/2**.
- Enables DHCP option 79.

```
switch(config)# dhcpv6-relay
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# vlan access 10
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vlan access 20
switch(config-if)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::22/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000 egress vlan 50
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-servers egress
vlan50
switch(config-if-vlan)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::21/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
```

```

switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000 egress vlan 50
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-servers egress
vlan50
switch(config-if-vlan)# exit
switch(config)# end

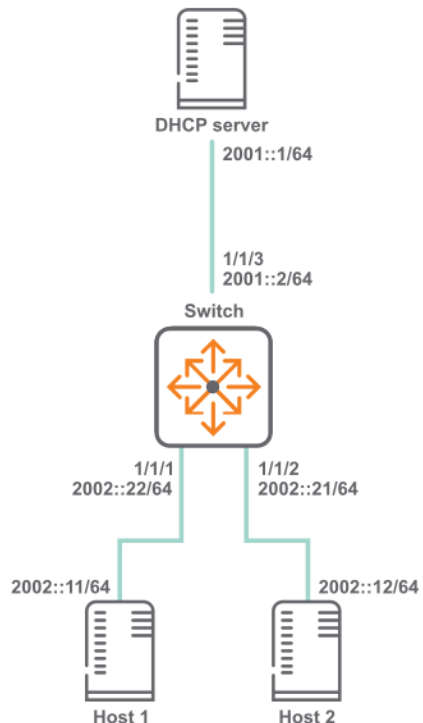
```

Use Case

This section explain the use cases for DHCPv6 relay agent.

DHCPv6 relay scenario 1

In this scenario, DHCP relay on the switch enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



Procedure

1. Enable DHCP relay.

```

switch# config
switch(config)# dhcpv6-relay

```

2. Define an IPv6 helper address.

- On interface `vlan 10` and `vlan 20`.

```

switch(config)# dhcpv6-relay
switch(config)# interface 1/1/1
switch(config)# no shutdown
switch(config-if)# vlan access 10
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config)# no shutdown
switch(config-if)# vlan access 20
switch(config-if)# exit

```

```

switch(config)# interface vlan 10
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::22/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000 egress
vlan30
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-servers
egress vlan50
switch(config-if-vlan)# exit
switch(config)# interface vlan 20
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ipv6 address 2002::21/64
switch(config-if-vlan)# ipv6 helper-address unicast 2001::1
switch(config-if-vlan)# ipv6 helper-address multicast ff01::1:1000 egress
vlan30
switch(config-if-vlan)# ipv6 helper-address multicast all-dhcp-servers
egress vlan50
switch(config-if-vlan)# exit
switch(config)# end

```

3. Verify DHCP relay configuration.

- When interface `vlan 10` and `vlan 20` is configured as relay.

```

switch# show dhcpv6-relay
  DHCPv6 Relay Agent : enabled
  Option 79          : disabled
switch# show ipv6 helper-address

IP Helper Addresses

Interface: vlan10
IP Helper Address          Egress Port
-----
all-dhcp-servers          vlan50
ff01::1:1000              vlan30
2001::1                    -

Interface: vlan20
IP Helper Address          Egress Port
-----
all-dhcp-servers          vlan50
ff01::1:1000              vlan30
2001::1                    -

```

DHCP relay (IPv6) commands

dhcpv6-relay

```

dhcpv6-relay [l2vpn-clients|source-interface]
no dhcpv6-relay [l2vpn-clients|source-interface]

```

Description

Enables DHCPv6 relay support. DHCPv6 relay is disabled by default. DHCP relay is not supported on the management interface. Best practices is to disable this configuration on all the VXLAN tunnel endpoints (VTEPs), to avoid forwarding duplicate DHCPv6 requests to the server.

The **no** form of this command disables DHCP relay support.

DHCPv6 Relay requires that you configure the egress interface using the [ipv6 helper-address](#) command. The egress interface of a VTEP is used as an underlay, so a DHCPv6 Relay Multicast ipv6 address is not supported in a VXLAN topology.

Parameter	Description
<code>l2vpn-clients</code>	Enables packets from l2vpn clients to be forwarded to configured servers. Enabled by default.
<code>source-interface</code>	Enables DHCPv6 relay to use the configured source interface.

Usage

In Asymmetric/Symmetric Integrated Routing and Bridging (IRB) VXLAN deployments with a VLAN extension in subset of VTEPs, client DHCPv6 broadcast requests are received by all the VTEPs where a client VLAN is configured. A DHCPv6 relay agent on those VTEPs forward DHCPv6 packets to configured DHCPv6 server(s). As DHCPv6 requests are forwarded by multiple DHCPv6 relay agents, the DHCPv6 server receives duplicate copies of the same packet. When this configuration is disabled, the DHCPv6 relay agent on VTEPs ignores DHCPv6 request packets that are received from client MACs addresses learned via EVPN.

Examples

Enables DHCPv6 relay support.

```
switch(config)# dhcpv6-relay
```

Removes DHCPv6 relay support.

```
switch(config)# no dhcpv6-relay
```

Command History

Release	Modification
10.12.1000	l2vpn-clients and source-interface added.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	<code>config</code>	Administrators or local user group members with execution rights for this command.

dhcpv6-relay option 79

```
dhcpv6-relay option 79  
no dhcpv6-relay option 79
```

Description

Enables support for DHCP relay option 79. When enabled, the DHCPv6 relay agent forwards the link-layer address of the client. This option is disabled by default.

The **no** form of this command disables support for DHCP relay option 79.

Examples

Enables DHCP option 79 support.

```
switch(config)# dhcpv6-relay option 79
```

Disables DHCP option 79 support.

```
switch(config)# no dhcpv6-relay option 79
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ipv6 helper-address

```
ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
```

```
no ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
```

```
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>} egress <PORT-  
NUM>
```

```
no ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>} egress <PORT-  
NUM>
```

Description

Defines the address of a remote DHCPv6 server or DHCPv6 relay agent. Up to eight addresses can be defined. The DHCPv6 agent forwards DHCPv6 client requests to all defined servers.

Not supported on the OOBM interface.

The **no** form of this command removes an IP helper address.

Parameter	Description
<UNICAST-IPV6-ADDR>	Specifies the unicast helper IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<MULTICAST-IPV6-ADDR>	Specifies the multicast helper IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Parameter	Description
<code>all-dhcp-servers</code>	Specifies all the DHCP server IPv6 addresses for the interface.
<code>egress <PORT-NUM></code>	Specifies the port number on which DHCPv6 service requests are relayed to a multicast destination. The egress port must be different than the one on which the multicast helper address is configured. Format: member/slot/port.
<code>vrf <VRF-NAME></code>	Specifies the name of the VRF from which the specified protocol sets its source IP address.

Examples

Defining a multicast IPv6 helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if)# ipv6 helper-address multicast 2001:DB8:0:0:0:0:0:1 egress 1/1/2
```

Removing the IP helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if)# no ipv6 helper-address multicast 2001:DB8:0:0:0:0:0:1 egress 1/1/2
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	<code>config-if</code>	Administrators or local user group members with execution rights for this command.

show dhcpv6-relay

```
show dhcpv6-relay
```

Description

Shows DHCP relay configuration settings.

Example

```
switch# show dhcpv6-relay
DHCPv6 Relay Agent : enabled
Option 79           : disabled
L2vpn-clients       : enabled
Source-interface    : enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	

show ipv6 helper-address

show ipv6 helper-address [interface <INTERFACE-ID>]

Description

Shows the helper IP addresses defined for all interfaces or a specific interface.

Parameter	Description
interface <INTERFACE-ID>	Specifies an interface. Format: member/slot/port .

Examples

```
switch# show ipv6 helper-address
```

```
Interface: 1/1/1
IPv6 Helper Address          Egress Port
-----
2001:db8:0:1::              -
FF01::1:1000                1/1/2
```

```
Interface: 1/1/2
IPv6 Helper Address          Egress Port
-----
2001:db8:0:1::              -
```

```
switch# show ipv6 helper-address interface 1/1/1
```

```
Interface: 1/1/1
IPv6 Helper Address          Egress Port
-----
2001:db8:0:1::              -
FF01::1:1000                1/1/2
```

```
switch# show ipv6 helper-address interface vlan20
```

```
Interface: vlan20
IP Helper Address            Egress Port
-----
2001::1                      -
ff01::1:1000                vlan30      default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	

FAQ

1. What ports use DHCP for switch assignments?

You can set up a DHCP switch with a management VRF port; if the management VRF port is unavailable, use an IN-Band data port.

2. Is DHCP assignment available on boot up from the factory?

Yes, DHCP assignment is available on boot up from the factory. This assists Zero Touch depending on the platform; this can be a management VRF or an IN-Band data port on the default VLAN 1.

3. Is DHCP assignment available on both management VRF and IN-Band data ports simultaneously?

Yes, DHCP assignment is possible on both management VRF and IN Band data ports, but it depends on the platform.

If both ports can get a DHCP address, the DHCP address from the management VRF port is used for Zero Touch Provisioning instead of the one from the IN-Band data port. The data ports will wait 30 seconds before taking ownership of Zero Touch Provisioning (ZTP).

4. Which VLAN is assigned a DHCP address?

The factory default VLAN to which the DHCP address is assigned is VLAN1.

Users can configure the VLAN for DHCP allocation on platforms supporting multiple VLAN configurations. Only a single VLAN can support DHCP assignments.

5. Can DHCP assignments be disabled in an HPE Aruba Networking AOS-CX switch?

Use the **no ip dhcp** command on the required VLAN interface or management VRF port you can disable the DHCP assignments in an HPE Aruba Networking AOS-CX switch.

A static configuration option is available on both ports.

6. Can IP DHCP switch assignment be carried out on a Routed only Port (RoP)?

No, RoP is not supported. Only VLAN and OOBM ports can be allocated an address with DHCP.

7. Can an HPE Aruba Networking AOS-CX switch support DHCP IPv6 Allocation?

Yes, an HPE Aruba Networking AOS-CX switch can support DHCP IPv6 allocation. However, for Zero Touch Provisioning (ZTP), IPv4 is required.

8. Do HPE Aruba Networking AOS-CX switches support DHCP server?

Since the software release of AOS-CX 10.10, all platforms support DHCP server, except for 6100, 6000, and 4100i.

9. Can HPE Aruba Networking AOS-CX switches support DHCP address exclusion from a subnet?

Not formally; however, you can use static bindings to unused MAC addresses as a workaround on a smaller scale.

10. Do all HPE Aruba Networking AOS-CX switches support DHCP helper?

Since the software release of AOS-CX 10.10, all platforms support DHCP helper for IPv4 and IPv6.

11. Can DHCP Server and Relay simultaneously be supported on a switch?

Yes, since the software release AOS-CX 10.08, the platforms that support DHCP Server and Relay both can be set up simultaneously.

12. How many helper addresses can be used?

The Switched Virtual Interface can use up to eight helper addresses for IPv4 and IPv6.

13. Is DHCP Relay supported for multi-netted addresses?

Yes, for both IPv4 and IPv6 DHCP Relay supports multi-netted addresses. The Gateway IP Address (GIADDR) used in the DHCP request will be the lowest IP address configured on the interface.

You can override the lowest IP address using the `bootp-gateway` command. IPV6 does not support bootp.

14. Can a multi-netted address support GIADDR from a different address?

Yes, for IPv4 only. For more information, see [DHCP relay agent](#).

15. Can the binding of DHCP and IP remain after a switch reboot?

Yes, the binding of DHCP and IP remains by using an external storage option `dhcpx4-snooping external-storage`.

16. If multiple 'ip-helper' IPs are configured for DHCP relay on an interface or SVI, which switch will forward the DHCP requests?

In case of multiple 'ip-helper' IPs configured for DHCP relay on an interface or SVI, the client will receive multiple offers from different DHCP servers, and the client will choose which one to accept.

17. What happens when global *dhcpx-smart-relay* is enabled in an interface or SVI with multiple IP addresses (multi-netting) that services DHCP clients in different subnets?

On enabling global **dhcpx-smart-relay** in an interface or SVI with multiple IP addresses (multi-netting) that services DHCP clients in different subnets, the AOS-CX device will first use the primary interface or SVI IP address as the unicast source to the DHCP server. If no response is received the AOS-CX device will use the secondary IPs from the lowest-numbered IP to the highest until a DHCP response is received.

18. What are the troubleshooting commands for DHCP relay agent?

The initial troubleshooting for DHCP relay agent can be done using the following commands:

- `show dhcp-relay`
- `show dhcp-relay bootp-gateway`
- `show ip helper-address`

Next level of troubleshooting can be done by taking packets captured at the DHCP server side and AOS-CX side. For more information about AOS-CX packet capture, see *Mirroring* chapter of *AOS-CX Monitoring Guide*.

Perform additional AOS-CX side DHCP debugging using **`debug dhcprelay`** and **`debug dhcpv6relay`** commands. For more information, see *Debug logging* chapter of *AOS-CX Diagnostics and Supportability Guide*.

Chapter 6

DHCP snooping

DHCP is a protocol used by DHCP servers in IP networks to dynamically allocate network configuration data to client devices (DHCP clients). Possible network configuration data includes user IP address, subnet mask, default gateway IP address, DNS server IP address, and lease duration. The DHCP protocol enables DHCP clients to be dynamically configured with such network configuration data without any manual setup process.

DHCP snooping is a security feature that helps avoid problems caused by an unauthorized DHCP server on the network that provides invalid configuration data to DHCP clients. A user without malicious intent may cause this problem by unknowingly adding to the network a switch or other device that includes a DHCP server enabled by default. In some cases, a user with malicious intent adds a DHCP server to the network as part of their Denial of Service or Man in the Middle attack.

DHCP snooping helps prevent such problems by distinguishing between trusted ports connected to legitimate DHCP servers and untrusted ports connected to general users. DHCP packets are forwarded between trusted ports without inspection. DHCP packets received on other switch ports are inspected before being forwarded. DHCP Packets from untrusted sources are dropped.

DHCP Snooping and DHCP relay can be configured on the same switch.

When DHCP snooping and DHCP relay are both enabled on a VLAN, the following actions occur:

- Received packet: DHCP snooping processes the DHCP packet before (possibly) handing it to DHCP relay.
 - Transmitted packet: DHCP packets sent by DHCP relay are intercepted by DHCP snooping to learn IP bindings.
-

For even more rigorous security that is applied in hardware on a packet-by-packet basis, you can use IP source lockdown feature as described in `IP source lockdown`.

DHCP server interoperation

DHCP server can not be configured with DHCP snooping.

DHCPv4 snooping conditions for dropping DHCPv4 packets

Applies only to DHCPv4 snooping.

Packet types that are dropped	Conditions for dropping the packets
DHCPOFFER, DHCPACK, DHCPNACK	▪ A packet from a DHCP server is received on an untrusted port. ▪ The switch is configured with a list of authorized DHCP server addresses and a DHCP response received on a trusted port, but the source IP address is not an authorized DHCP server.

Packet types that are dropped	Conditions for dropping the packets
DHCPRELEASE, DHCPDECLINE	A broadcast packet that has a MAC address in the DHCP binding database, but the port in the DHCP binding database does not match the port on which the packet is received.
All DHCP packet types	- A DHCP packet received on an untrusted port in which the DHCP client hardware MAC address does not match the source MAC address in the packet. - A DHCP packet containing DHCP relay information (option 82) is received on an untrusted port.

Protocol details

1. When a port is configured as a trusted port, all the dynamic IP binding entries learned on that port will be deleted.
2. When a client is connected on a trusted port, the dynamic IP binding entries will not be learned on the switch, even though the client gets an IP address.
3. If DHCPv4 snooping is enabled on two back-to-back access switches, DHCP packets will be dropped. Since by default option 82 is enabled on DHCPv4 snooping and the default policy is drop. The second switch with DHCPv4 snooping enabled drops the packets. In this scenario the user should enable DHCPv4 snooping option 82 on one switch, or else you can disable on both.

Supported platform and standards

The following table list the supported platforms for DHCP snooping.

Table 1: *Support platforms for DHCP snooping*

Platform	DHCP snooping
4100i	Yes
6000	Yes
6100	Yes

Table 2: *Scale*

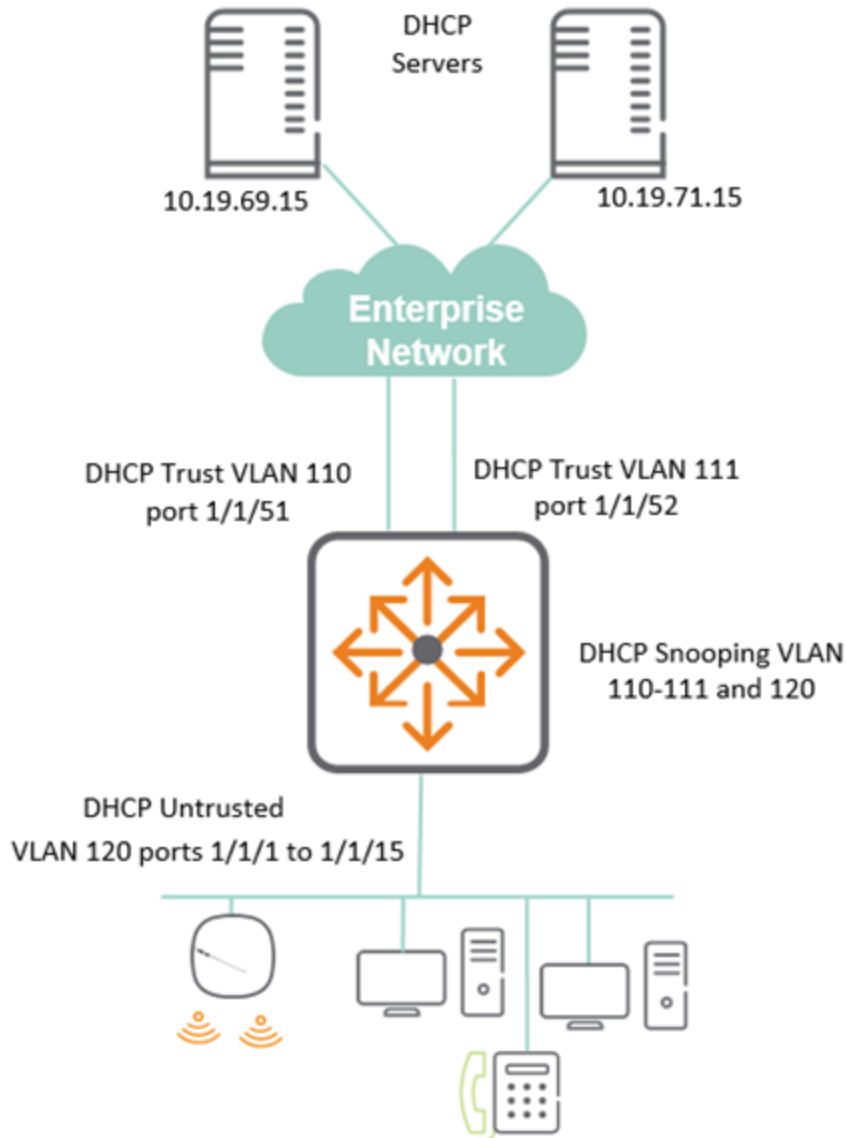
Platform	IP Bindings per port	IP Bindings per device
4100i	256	256
6000	256	256
6100	256	256

DHCPv4 Snooping Use case

DHCP Snooping is used to protect from rogue or unwanted attacks from posing DHCP services.

In the following use case:

- Set the switch for DHCPv4 snooping.
- Allow DHCP communication from the uplinks in the Enterprise network port 1/1/51 and 1/1/52.
- For communication to occur, identify the authorized DHCP servers 10.19.69.15 and 10.19.71.15.
- LAN facing user ports 1/1/1 to 1/1/15 is not configured as trusted.



Key configuration portion for above set up is shown with the relevant DHCPv4 snooping parameters:

```
dhcpv4-snooping                                <--enable DHCPv4 snooping
dhcpv4-snooping authorized-server 10.19.69.15   <--allow authorized servers
dhcpv4-snooping authorized-server 10.19.71.15
vlan 110
  description uplink
  dhcpv4-snooping                               <--enable DHCPv4 snooping
on required VLANs
vlan 111
  description uplink
```

```

    dhcpv4-snooping
vlan 120
    description User
    dhcpv4-snooping
interface 1/1/1
    no shutdown
    description User
    vlan access 120

...

interface 1/1/15
    no shutdown
    description User
    vlan access 120
interface 1/1/51                                     <--Unlinks on 1/1/51 and 52 trusted
    no shutdown
    vlan trunk native 1
    vlan trunk allowed 110
    dhcpv4-snooping trust
interface 1/1/52
    vlan trunk native 1
    vlan trunk allowed 111
    dhcpv4-snooping trust
interface vlan 110
    description uplink
    ip address 172.16.69.1/30
interface vlan 111
    description uplink
    ip address 172.16.71.1/30
interface vlan 120
    description User
    ip address 172.16.10.1/254
    ip helper-address 10.19.69.15
    ip helper-address 10.19.71.15
...
ip route 0.0.0.0/0 172.16.69.2 distance 10
ip route 0.0.0.0/0 172.16.71.2 distance 20
...

```

Using the show dhcpv4-snooping command, you can see some of the following DHCPv4 information:

- Applied VLANs for snooping, 110-112 and 120
- The untrusted policy is set to drop packets
- The list of authorized DHCP servers are set to 10.19.69.15 and 10.19.71.15
- Set the trusted or untrusted ports and any DHCP bindings for those ports. Port 1/1/1 and 1/1/2 offer DHCP.

```

show dhcpv4-snooping

DHCPv4-Snooping Information

DHCPv4-Snooping      : Yes          Verify MAC Address   : Yes
Allow Overwrite Binding : No          Enabled VLANs        : 110-111,120
IP Binding Disabled VLANs :
Static Attributes     : No
Client Event Logs      : No

Option 82 Configurations

```

```
Untrusted Policy      : drop          Insertion           : Yes
Option 82 Remote-id   : mac
```

External Storage Information

```
Volume Name          : --
File Name             : --
Inactive Since       : --
Error                 : --
```

Flash Storage Information

```
File Write Delay : --
```

```
Active Storage : --
```

Authorized Server Configurations

VRF	Authorized Servers
default	10.19.69.15
default	10.19.71.15

Port Information

Port	Trust	Max Bindings	Static Bindings	Dynamic Bindings
1/1/51	Yes	0	0	0
1/1/52	Yes	0	0	0
1/1/1	No	1024	0	1
1/1/2	No	1024	0	1
!				
!output omitted				
!				
1/1/14	No	1024	0	0
1/1/15	No	1024	0	0

DHCP snooping commands

clear dhcp-snooping binding

```
clear dhcp-snooping binding {all | ip <IP-ADDR> vlan <VLAN-ID> | port <PORT-NUM> | vlan <VLAN-ID>}
```

Description

Clears DHCP snooping binding entries.

Parameter	Description
all	Specifies that all DHCP binding information is to be cleared.
ip <IP-ADDR> vlan <VLAN-ID>	Specifies the IP address and VLAN for which all DHCP binding information is to be cleared.

Parameter	Description
port <PORT-NUM>	Specifies the port number for which all DHCP binding information is to be cleared.
vlan <VLAN-ID>	Specifies the VLAN for which all DHCP binding information is to be cleared.

Examples

Clearing all DHCP binding information for IP address 192.168.2.4 and VLAN 5:

```
switch(config)# clear dhcp-snooping binding ip 192.168.2.4 vlan 5
```

Clearing all DHCP binding information for port 1/1/1:

```
switch(config)# clear dhcp-snooping binding port 1/1/1
```

Clearing all DHCP binding information for VLAN 10:

```
switch(config)# clear dhcp-snooping binding vlan 10
```

Clearing all DHCP binding information:

```
switch(config)# clear dhcp-snooping binding all
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping . The ipv4 parameter is deprecated and replaced with ip .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

clear dhcp-snooping statistics

```
clear dhcp-snooping statistics
```

Description

Clears all DHCP snooping statistics.

Examples

Clear all DHCP snooping statistics:

```
switch# clear dhcp-snooping statistics
```

Command History

Release	Modification
10.14	The dhcpx4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

dhcp-snooping

dhcp-snooping
no dhcp-snooping

Description

Enables DHCP snooping. DHCP snooping is disabled by default. DHCP snooping is not supported on the management interface.

The **no** form of the command disables DHCP snooping, flushing all the IP bindings learned since DHCP snooping was enabled.

Examples

Enabling DHCP snooping:

```
switch(config) # dhcp-snooping
```

Disabling DHCP snooping:

```
switch(config) # no dhcp-snooping
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping (in config-vlan context)

dhcp-snooping
no dhcp-snooping

Description

Enables DHCP snooping for the specified VLAN in the **config-vlan** context. DHCP snooping is disabled by default for all VLANs.

The no form of the command disables DHCP snooping on the specified VLAN, flushing all the IP bindings learned for this VLAN since DHCP snooping was enabled for this VLAN.

Examples

Enabling DHCP snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcp-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCP snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcp-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-vlan	Administrators or local user group members with execution rights for this command.

dhcp-snooping allow-overwrite-binding

```
dhcp-snooping allow-overwrite-binding
no dhcp-snooping allow-overwrite-binding
```

Description

Allows binding to be overwritten for the same IP address. When enabled, and a DHCP server offers a host an IP address that is already bound to an existing host in the binding table, the existing binding is overwritten for the new host if the new host is successfully able to acquire the same IP address. This overwriting is disabled by default, causing the DHCP server offers to be dropped.

The **no** form of the command disables DHCP snooping overwrite binding.

Examples

Enabling DHCP snooping overwrite binding:

```
switch(config) # dhcp-snooping allow-overwrite-binding
```

Disabling DHCP snooping overwrite binding:

```
switch(config) # no dhcp-snooping allow-overwrite-binding
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping authorized-server

```
dhcp-snooping authorized-server <IP-ADDR> [vrf <VRF-NAME>]
no dhcp-snooping authorized-server <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Adds an authorized (trusted) DHCP server to a list of authorized servers for use by DHCP snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCP servers are considered to be trusted for DHCP snooping purposes.

The **mgmt** VRF cannot be used with this command.

The no form of this command deletes the specified DHCP server from the authorized list.

Parameter	Description
<IP-ADDR>	Specifies the IP address of the trusted DHCP server.
vrf <VRF-NAME>	Specifies the VRF name.

Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

Examples

Adding DHCP servers 192.168.2.2, 192.168.2.3, and 192.168.2.10 to the authorized server list:

```
switch(config)# dhcp-snooping authorized-server 192.168.2.2
switch(config)# dhcp-snooping authorized-server 192.168.2.3 vrf default
switch(config)# dhcp-snooping authorized-server 192.168.2.10 vrf default
```

Removing DHCP server 192.168.2.3 from the authorized server list:

```
switch(config)# no dhcp-snooping authorized-server 192.168.2.3 vrf default
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.09	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping event-log client

dhcp-snooping event-log client
no dhcp-snooping event-log client

Description

This command enables or disables dhcp-snooping client level event logs that help with client telemetry on a remote management station such as HPE Aruba Networking Central. By default, client level event logs are disabled. The **no** form of this command disables client-level event logs for DHCP snooping after they are enabled. View these logged DHCP snooping events by issuing the command **show events -c dhcp-snooping**.

For additional information on DHCP-related event logging, please refer to the Event Log Message Reference Guide.

Examples

Enabling DHCP client level event logs:

```
switch(config)# dhcp-snooping event-log client
```

Disabling external storage:

```
switch(config)# no dhcp-snooping event-log client
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping external-storage

```
dhcp-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
no dhcp-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

Description

Configures external storage to be used for backing up IP bindings (used by DHCP snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IP bindings from the configured external storage file to populate its local cache.

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in an external storage file.

Parameter	Description
volume <VOL-NAME>	Specifies the name of the existing external storage volume where the IP bindings file will be saved. Before running the dhcp-snooping external-storage volume command, first create the external storage volume using command external-storage <VOLUME-NAME> . See <i>External storage commands</i> in the <i>Command-Line Interface Guide</i> .
file <FILE-NAME>	Specifies the file name to use for storing IP bindings. Maximum 255 characters.

Configuring IP bindings storage in file **dsnoop_ipbindings** on existing volume **dhcp_snoop**:

```
switch(config)# dhcp-snooping external-storage volume dhcp_snoop file dsnoop_ipbindings
```

Disabling external storage:

```
switch(config)# no dhcp-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcp-snooping external-storage volume dhcp_snoop
dhcp-snooping will use flash storage to store IP Binding database
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping flash-storage

```
dhcp-snooping flash-storage [delay <DELAY>]  
no dhcp-snooping flash-storage [delay <DELAY>]
```

Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCP snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting **delay <DELAY>** sets the default delay of 900 seconds.

To reduce switch flash aging it is recommended that you use external storage (command **dhcp-snooping external-storage**) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The **no** form of this command disables the saving of IP bindings in flash storage.

Parameter	Description
delay <DELAY>	Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400.

Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcp-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to use an
external-storage.
Do you want to continue (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcp-snooping flash-storage
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping max-bindings

```
dhcp-snooping max-bindings <MAX-BINDINGS>
no dhcp-snooping max-bindings <MAX-BINDINGS>
```

Description

Sets the maximum number of DHCP bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max binding is the maximum value of the range.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<MAX-BINDINGS>	Specifies the maximum number of DHCP bindings. Range: 1 to 256.

Examples

Set the DHCP max bindings to 256 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# dhcp-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Revert DHCP max bindings to its default on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no dhcp-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

dhcp-snooping option 82

```
dhcp-snooping option 82 [remote-id {mac | subnet-ip}][untrusted-policy {drop | keep | replace}]
no dhcp-snooping option 82 [remote-id {mac | subnet-ip}][untrusted-policy {drop | keep | replace}]
```

Description

Configures the addition of option 82 DHCP relay information to DHCP client packets that are being forwarded on trusted ports. DHCP relay is enabled by default.

In the switch default state and when this command is entered without parameters (**dhcp-snooping option 82**), this default configuration is used:

`dhcp-snooping option 82 remote-id mac untrusted-policy drop`

When **remote-id** is omitted, its default (**mac**) is used. When **untrusted-policy** is omitted, its default (**drop**) is used.

The no form of this command disables DHCP snooping option 82.

Parameter	Description
<code>remote-id</code>	Specifies what address to use as the remote ID for the <code>replace</code> option of <code>untrusted-policy</code> . Specify one of these address types:
<code>mac</code>	The default. Uses the switch MAC address as the remote ID.
<code>subnet-ip</code>	Uses the IP address of the client VLAN as the remote ID.
<code>untrusted-policy</code>	Specifies what action to take for DHCP packets (with option 82) that are received on untrusted ports. Specify one of these actions:
<code>drop</code>	The default. Drop DHCP packets (with option 82) without forwarding them.
<code>keep</code>	Forward DHCP packets (with option 82).
<code>replace</code>	Replace the option 82 information in the DHCP packets with whatever is set for remote-id (one of: mac , subnet-ip , or mgmt-ip) and forward the packets.

Examples

Configuring DHCP snooping option 82 with the keep action:

```
switch(config)# dhcp-snooping option 82 untrusted-policy keep
```

Disabling DHCP snooping option 82:

```
switch(config)# no dhcp-snooping option 82 untrusted-policy keep
```

Command History

Release	Modification
10.14	The dhcipv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping static-attributes

dhcp-snooping static-attributes
no dhcp-snooping static-attributes

Description

Enables storage of static attributes provided to the DHCP client by DHCP server during DHCP packet exchange. Disabled by default. When enabled, the following attributes are stored in OVSDb along with the client IP binding entry:

1. Name server IP addresses: DNS server IPs provided by the DHCP server to the client. Maximum: 3 per client.
2. Default gateway IP address: Router IP addresses provided by DHCP server to the client. Maximum: 3 per client.
3. Server IP address: IP address of the DHCP server that leased the IP to the client.

The **no** form of the command disables storing of client static attributes. After disabling, existing client static attributes will be flushed.

Examples

Enabling the storage of DHCP snooping static attributes:

```
switch(config)# dhcp-snooping static-attributes
```

Disabling the storage of DHCP snooping static attributes:

```
switch(config)# no dhcp-snooping static-attributes
```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcp-snooping trust

```
dhcp-snooping trust
no dhcp-snooping trust
```

Description

Enables DHCP snooping trust on the selected port. Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The **no** form of the command disables DHCP snooping trust on the selected port.

Examples

Enabling DHCP snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcp-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCP snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcp-snooping trust
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.14	The dhcpx4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

dhcp-snooping verify mac

```
dhcp-snooping verify mac
no dhcp-snooping verify mac
```

Description

This command enables verification of the hardware address field in DHCP client packets. When enabled, the DHCP client hardware address field and the source MAC address must be the same for packets received on untrusted ports or else the packet is dropped. This DHCP snooping MAC verification is enabled by default.

The **no** form of the command disables DHCP snooping MAC verification.

Examples

Enabling DHCP snooping MAC verification:

```
switch(config)# dhcp-snooping verify mac
```

Disabling DHCP snooping MAC verification:

```
switch(config)# no dhcp-snooping verify mac
```

Command History

Release	Modification
10.14	The dhcpx4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show dhcp-snooping

```
show dhcp-snooping
```

Description

Shows the DHCP snooping configuration.

Examples

Showing the DHCP snooping configuration:

```
switch# show dhcp-snooping  
  
dhcp-snooping Information
```

```

dhcp-snooping           : Yes           Verify MAC Address    : Yes
Allow Overwrite Binding : No           Enabled VLANs         : 1-100
Static Attributes       : Yes
Client Event Logs       : Yes

```

Option 82 Configurations

```

Untrusted Policy        : replace       Insertion              : Yes
Option 82 Remote-id     : mac

```

External Storage Information

```

Volume Name      : ipbinding
File Name        : ipv4Bindings
Inactive Since   : 01:23:20 09/10/2021
Error            : File Write Failure

```

Flash Storage Information

```

File Write Delay : 300 seconds
Active Storage   : External

```

Authorized Server Configurations

VRF	Authorized Servers
default	1.1.10.3
default	10.10.10.1
default	10.10.10.56
default	200.10.10.3
green	1.1.10.3
green	1.10.10.3
green	10.10.100.3
red	192.168.122.53
red	192.168.122.121

Port Information

Port	Trust	Max Bindings	Static Bindings	Dynamic Bindings
1/1/2	Yes	5000	50	0
1/1/3	Yes	8192	0	0
1/1/5	Yes	8192	0	22
1/1/16	No	100	0	0
10/10/10	No	8100	320	200
lag120	No	512	0	0

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.

Release	Modification
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show dhcp-snooping binding

show dhcp-snooping binding

Description

Shows the DHCP snooping binding configuration.

Parameter	Description
detail	Shows detailed information for active IP bindings on the system.

Examples

Showing the DHCP snooping binding configuration:

```
switch(config)# show dhcp-snooping binding
```

MacAddress	IP	VLAN	Interface	Time-Left
aa:b1:c1:dd:ee:ff	10.2.3.4	1	1/1/2	582
aa:b2:c2:dd:ee:ff	10.2.3.5	1	1/1/2	584

Showing detailed information for active IP bindings:

```
switch(config)# show dhcp-snooping binding detail
```

VLAN Id : 2, MAC : 00:50:56:96:74:46

IP	Interface	Time-Left
100.1.2.100	1/1/23	194

Static Attributes:

Default Router : 100.1.2.1, 192.1.1.1, 1.1.1.2

Server IP : 10.1.84.2

Name Servers : 192.1.1.2, 2.2.2.2, 1.1.1.1

VLAN Id : 3, MAC : 00:50:56:96:e5:8e

```

IP                Interface  Time-Left
-----
100.1.3.100       2/1/22    145

Static Attributes:
Default Router   : 100.1.3.1, 192.1.1.1, 1.1.1.2
Server IP       : 10.1.84.2
Name Servers     : 192.1.1.2, 2.2.2.2, 1.1.1.1

VLAN Id : 3, MAC : 00:11:01:00:00:03

IP                Interface  Time-Left
-----
100.1.3.99        2/1/24    137

Static Attributes:
Default Router   : 100.1.3.1, 192.1.1.1, 1.1.1.2
Server IP       : 10.1.84.2
Name Servers     : 192.168.0.1, 192.168.1.1, 192.168.2.1

```

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.10	Detail parameter added.
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show dhcp-snooping statistics

`show dhcp-snooping statistics`

Description

Shows the DHCP snooping statistics.

Examples

Showing the DHCP snooping statistics:

```
switch(config)# show dhcp-snooping statistics
```

Packet-Type	Action	Reason	Count
server	forward	from trusted port	5425
client	forward	to trusted port	3895
server	drop	received on untrusted port	117
server	drop	unauthorized server	214
client	drop	destination on untrusted port	78
client	drop	untrusted option 82 field	85
client	drop	bad DHCP release request	0
client	drop	failed verify MAC check	5
client	drop	failed on max-binding limit	15

Command History

Release	Modification
10.14	The dhcpv4-snooping keyword is deprecated and replaced with dhcp-snooping .
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

DHCPv6 snooping commands

clear dhcpv6-snooping binding

```
clear dhcpv6-snooping binding {all | ip <IPV6-ADDR> vlan <VLAN-ID> | interface <IFNAME> |
vlan <VLAN-ID>}
```

Description

Clears DHCPv6 snooping binding entries.

Parameter	Description
all	Specifies that all DHCPv6 binding information is to be cleared.
ip <IPV6-ADDR> vlan <VLAN-ID>	Specifies the IPv6 address and VLAN for which all DHCPv6 binding information is to be cleared.
interface <IFNAME>	Specifies the interface for which all DHCPv6 binding information is to be cleared.

Parameter	Description
vlan <VLAN-ID>	Specifies the VLAN for which all DHCPv6 binding information is to be cleared. Range: 1 to 4094.

Examples

Clearing all DHCPv6 binding information for 5000::1 vlan 1:

```
switch(config)# clear dhcpv6-snooping binding ip 5000::1 vlan 1
```

Clearing all DHCPv6 binding information for interface 1/1/10:

```
switch(config)# clear dhcpv6-snooping binding interface 1/1/10
```

Clearing all DHCPv6 binding information for VLAN 10:

```
switch(config)# clear dhcpv6-snooping binding vlan 10
```

Clearing all DHCPv6 binding information:

```
switch(config)# clear dhcpv6-snooping binding all
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

clear dhcpv6-snooping statistics

```
clear dhcpv6-snooping statistics
```

Description

Clears all DHCPv6 snooping statistics.

Examples

Clear all DHCPv6 snooping statistics:

```
switch# clear dhcpv6-snooping statistics
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

dhcpv6-snooping

dhcpv6-snooping
no dhcpv6-snooping

Description

Enables DHCPv6 snooping. DHCPv6 snooping is disabled by default. DHCPv6 snooping is not supported on the management interface.

The no form of the command disables DHCPv6 snooping, flushing all the IP bindings learned since DHCPv6 snooping was enabled.

Examples

Enabling DHCPv6 snooping:

```
switch(config)# dhcpv6-snooping
```

Disabling DHCPv6 snooping:

```
switch(config)# no dhcpv6-snooping
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping (in config-vlan context)

dhcpv6-snooping
no dhcpv6-snooping

Description

Enables DHCPv6 snooping in the `config-vlan` context. DHCPv6 snooping is disabled by default for all VLANs.

The `no` form of the command disables DHCPv6 snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since DHCPv6 snooping was enabled for this VLAN.

Examples

Enabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-vlan	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping authorized-server

dhcpv6-snooping authorized-server <IPv6-ADDR> [vrf <VRF-NAME>]

```
no dhcpv6-snooping authorized-server <IPv6-ADDR> [vrf <VRF-NAME>]
```

Description

Adds an authorized (trusted) DHCPv6 server to a list of authorized servers for use by DHCPv6 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCPv6 servers are considered to be trusted for DHCPv6 snooping purposes.

The `mgmt` VRF cannot be used with this command.

Configure the link local IPv6 address instead of global IPv6 address of the DHCPv6 server as the authorized-server. For example:

```
switch(config)# dhcpv6-snooping authorized-server fe80::2ca4:fa40:d4cd:bc2f
```

The `no` form of this command deletes the specified DHCPv6 server from the authorized list.

Parameter	Description
<IPv6-ADDR>	Specifies the IPv6 address of the trusted DHCPv6 server.
vrf <VRF-NAME>	Specifies the VRF name.

Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

Examples

Adding DHCP servers ABCD:5ACD::2000, and ABCD:5ACD::2010 to the authorized server list:

```
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf default
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2010 vrf default
```

Removing DHCP server ABCD:5ACD::2000 from the authorized server list:

```
switch(config)# no dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf default
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping event-log client

```
dhcpv6-snooping event-log client
no dhcpv6-snooping event-log client
```

Description

This command enables or disables DHCPv6 snooping client level event logs that help with client telemetry on a remote management station such as HPE Aruba Networking Central. By default, client level event logs are disabled. The **no** form of this command disables client-level event logs for DHCPv6 snooping after they are enabled. View these logged DHCPv6 snooping events by issuing the command `show events -c dhcpv6-snooping`.

For additional information on DHCP-related event logging, please refer to the Event Log Message Reference Guide.

Examples

Enabling DHCPv6 client level event logs:

```
switch(config)# # dhcpv6-snooping event-log client
```

Disabling external storage:

```
witch(config)# # no dhcpv6-snooping event-log client
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping external-storage

```
dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
no dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

Description

Configures external storage to be used for backing up IPv6 bindings (used by DHCPv6 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IPv6 bindings from the configured external storage file to populate its local cache.

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IPv6 bindings in an external storage file.

Parameter	Description
volume <VOL-NAME>	Specifies the name of the existing external storage volume where the IPv6 bindings file will be saved. Before running the <code>dhcpv6-snooping external-storage volume</code> command, first create the external storage volume using command <code>external-storage <VOLUME-NAME></code> . See <i>External storage commands</i> in the <i>Command-Line Interface Guide</i> .
file <FILE-NAME>	Specifies the file name to use for storing IPv6 bindings. Maximum 255 characters.

Examples

Configuring IPv6 bindings storage in file `ipv6Bindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv6-snooping external-storage volume dhcp_snoop file
ipv6Bindings
```

Disabling external storage:

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
DHCPv6-Snooping will use flash storage to store IP Binding database
switch(config)#
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping flash-storage

```
dhcpv6-snooping flash-storage [delay <DELAY>]
no dhcpv6-snooping flash-storage [delay <DELAY>]
```

Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCPv6 snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting `delay <DELAY>` sets the default delay of 900 seconds.

To reduce switch flash aging it is recommended that you use external storage (command `dhcpv6-snooping external-storage`) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.

When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The `no` form of this command disables the saving of IP bindings in flash storage.

Parameter	Description
<code>delay <DELAY></code>	Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400.

Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcpv6-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to use an
external-storage.
Do you want to continue (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcpv6-snooping flash-storage
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping max-bindings

```
dhcpv6-snooping max-bindings <MAX-BINDINGS>  
no dhcpv6-snooping max-bindings <MAX-BINDINGS>
```

Description

Sets the maximum number of DHCPv6 bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max binding is the maximum value of the range.

The no form of the command reverts max bindings for the selected interface to its default.

Parameter	Description
<MAX-BINDINGS>	Specifies the maximum number of DHCP bindings. Range: 1 to 256.

Examples

Set the DHCPv6 max bindings to 256 on interface 1/1/1:

```
switch(config)# interface 1/1/1  
switch(config-if)# dhcpv6-snooping max-bindings 256  
switch(config-if)# exit  
switch(config)#
```

Revert DHCPv6 max bindings to its default on interface 1/1/1:

```
switch(config)# interface 1/1/1  
switch(config-if)# no dhcpv6-snooping max-bindings 256
```

```
switch(config-if) # exit  
switch(config) #
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

dhcpv6-snooping trust

```
dhcpv6-snooping trust  
no dhcpv6-snooping trust
```

Description

Enables DHCPv6 snooping trust on the selected interface. Only server packets received on trusted interfaces are forwarded. All the interfaces are untrusted by default.

The no form of the command disables DHCPv6 snooping trust on the selected interface.

config-if

Examples

Enabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config) # interface 2/2/1  
switch(config-if) # dhcpv6-snooping trust  
switch(config-if) # exit  
switch(config) #
```

Disabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config) # interface 2/2/1  
switch(config-if) # no dhcpv6-snooping trust  
switch(config-if) # exit  
switch(config) #
```

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.

Release	Modification
10.08	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show dhcpv6-snooping

show dhcpv6-snooping

Description

Shows the DHCPv6 snooping configuration.

Examples

Showing the DHCPv6 snooping configuration:

```
switch# show dhcpv6-snooping

DHCPv6-Snooping Information

DHCPv6-Snooping      : Yes   Enabled VLANs      : 1,5,7,100-110
Trusted Port Bindings Enabled VLANs :
Client Event Logs    : Yes

External Storage Information

Volume Name          : dhcp_snoop
File Name             : ip_binding
Inactive Since       : 01:23:20 09/10/2021
Error                 : Failed to write external storage

Flash Storage Information
File Write Delay      : 300 seconds
Active Storage        : External

Authorized Server Configurations
VRF                   Authorized Servers
-----
default
2001:0db8:85a3:0000:0000:8a2e:0370:7334
default              2002::2
default              2004::1
red                  2002::1
red                  2002::2
red                  2002::9
green                5000::1
green                5000::2
green                5000::3
green                5000::7
green                5000::8

Port Information
```


Port	Trust	Max Bindings	Static Bindings	Dynamic Bindings
1/1/2	Yes	0	0	0
1/1/3	Yes	0	3	0
1/1/5	Yes	0	22	0
1/1/16	No	256	0	20
10/10/10	No	256	12	7
lag120	No	256	3	0

Command History

Release	Modification
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.08	Updated example with flash storage information.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show dhcpv6-snooping binding

show dhcpv6-snooping binding

Description

Shows the DHCPv6 snooping binding configuration.

Examples

Showing the DHCPv6 snooping binding configuration:

```
switch# show dhcpv6-snooping binding

IP Binding Information
=====
MAC-ADDRESS      IPV6-ADDRESS      VLAN  INTERFACE
TIME-LEFT
-----
00:50:56:96:e4:cf  aaaa:bbbb:cccc:dddd:eeee:1234:5678:abcd  1      1/1/1
584
00:50:56:96:04:4d  1000::3          134     1/1/2
435
00:50:56:96:d8:3d  2000:1000::4     2002    lag123
21234
```

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show dhcpv6-snooping statistics

show dhcpv6-snooping statistics

Description

Shows the DHCPv6 snooping statistics.

Examples

Showing the DHCPv6 snooping statistics:

```
switch(config)# show dhcpv6-snooping statistics
```

Packet-Type	Action	Reason	Count
server	forward	from trusted port	12
client	forward	to trusted port	20
server	drop	received on untrusted port	5
server	drop	unauthorized server	4
client	drop	destination on untrusted port	2
client	drop	bad DHCP release request	5
server	drop	relay reply on untrusted port	2
client	drop	failed on max-binding limit	5

Command History

Release	Modification
10.09.1000	Command introduced for the 8360 Switch Series.
10.09	Command introduced for the 6000 and 6100 Switch Series.
10.07 or earlier	Command introduced for the 4100i Switch Series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

Troubleshooting

DHCP client not receiving IP

1. Trusted port configuration

Check if the server reachable port is configured as trusted.

2. Validate Packet drop counters

Use `show dhcpv4-snooping statistics` and `show dhcpv6-snooping statistics` commands to view the packet drop counters. This helps to identify a possible reason for the termination of DHCP packet exchange between client and server. A few possible reasons are listed below:

- Maximum binding limit reached
 - System-wide limit—The maximum binding limit is shared between DHCPv4-Snooping, DHCPv6-Snooping, ND-Snooping, and IP-Source-Bindings. On reaching the limit, DHCP client requests get dropped. To check the limit, use the `show capacities ip-bindings` command.
 - Per port limit: This is a per port configurable value for a protocol. On reaching the limit, the subsequent client requests are dropped.
- Conflicts with IP-Source-Bindings—User-configured IP-Source-Bindings are given priority over dynamically learned bindings. Validate if the dynamic client is conflicting with any of the existing IP-Source-Binding.
- Authorized server check failures—Authorized server configuration is global. When one or more authorized servers are set up, all server replies should come from one of the authorized server IP addresses.
- CoPP packet drops—If the rate of the DHCP packets entering is higher than the configured value, excessive packets are dropped by CoPP. Check CoPP statistics for the DHCP class and configure the value accordingly. Higher CoPP values could cause other protocols to misbehave depending on the platform.

3. Server configuration and reachability

Validate the following to ensure that the DHCP Server is reachable and configured correctly.

- a. Check that the server is reachable over one of the trusted ports.
 - b. Check the server logs to understand the DHCP exchange status between client and server.
 - c. Check if DHCP Server pools are exhausted.
- #### 4. Option-82 configuration issues (Specific to DHCP Snooping IPv4)
- a. This configuration usually arises when DHCP Snooping and Relay are configured on the same VLAN. Option 82 at DHCP Snooping IPv4 should be disabled when configured for Inter-VRF DHCP-Relay. Use `no dhcpv4-snooping option 82` command to disable the functionality.
 - b. Option 82 is enabled by default, and the drop policy is set. If client packets are received with option 82, then they are dropped with this configuration.

5. Server and client are connected to same the VLAN
 - a. Check if the server is reachable on the same VLAN as the client.
 - b. If the server is on a different L3 network, configure DHCP-Relay to forward the packets between the client and server.
6. Authorized server configurations—When configured, the server assigning IP address should be one of the IP addresses from the authorized server list.

In DHCP snooping IPv6, if DHCP Relay is between a test device and a server, the authorised server configuration should include a link to the local IP address of the relay interface. This is because, while forwarding the server's response to the client, it will stamp a link to the local IP address as the source IP of the packet.

IP-Binding mismatch in VSX topologies

1. Both primary and secondary VSX peers should have the same clock value. If not, there is a possibility of inconsistent ip-bindings that could result in ip-binding sync in a loop. There is no straight-forward way to check the inconsistencies but by looking at the number of bindings and debug log analysis
2. Ensure the ISL state is up and operational.

DHCP client IP traffic is dropped at data path

1. This can happen when IP-Source-Lockdown is used along with DHCP snooping. Ensure IP-Bindings are learned for clients that are experiencing traffic failure issues
2. Check the hardware programming state of the IP-Binding entry by IP-Source-Lockdown. Use `show ipv4 source-lockdown bindings` or `show ipv6 source-lockdown bindings` commands.

A number of deployments provide access to WAN or public network only through a proxy server for the network security. This section provides information about the various methods of HTTP proxy configuration on the switch. The HTTP proxy configured is used to connect to HPE Aruba Networking Central as per the zero-touch provisioning requirement.

HTTP proxy location can be configured using the CLI or REST interface or auto-configured through the DHCP server connected to the switch.

- There are three sources for HTTP proxy location:
 - User configured HTTP proxy via CLI or REST interface.
 - DHCP options received via management VRF port.
 - DHCP options received via VLAN 1 on supported switch platforms.
- Operational configuration for HTTP proxy location is determined by the source with the highest priority. Source priority:
 1. User configured.
 2. DHCP options received via management VRF port.
 3. DHCP options received via VLAN 1.

-
- HTTP proxy location can only be a FQDN or an IPV4 address.
 - When HTTP proxy location and VRF are configured, they override any existing HTTP proxy location and VRF.
 - If this command is executed without the VRF parameter, the default VRF will be used.
 - Port number may need to be specified at the end of the IP address for FQDN to connect via HTTP proxy.
 - For example, 8088 is the TCP port number: http-proxy 192.168.248.248:**8088**
-

DHCP options commands

http-proxy

```
http-proxy {<FQDN | IPV4-ADDR> | IPV6-ADDR[:PORT]} [vrf <VRF-NAME>]  
no http-proxy [<FQDN | IPV4-ADDR>] [vrf <VRF-NAME>]
```

Description

Specifies HTTP proxy location and VRF.

When HTTP proxy location and VRF are configured on the switch, it overrides any existing HTTP proxy location and VRF as this has the highest priority over the values obtained from other sources.

Following locations can be used for the HTTP proxy location:

- A fully qualified domain name (FQDN).
- An IPv4 address with colon separated port number
- An IPv6 address with colon separated port number

When configuring an IPv6 address with a port number, the address must be specified inside square brackets. An example - [2001:0db8:85a3:0000:0000:8a2e:0370:7334]:8080.

If the command is entered without the VRF parameter, then the VRF used will be 'default' VRF. The **no** form of this command removes a specified HTTP proxy location.

Parameter	Description
<FQDN>	Specifies FQDN for HTTP proxy location.
<IPv4-ADDR>	Specifies IPV4 address for HTTP proxy location.
<IPv6-ADDR>	Specifies IPV6 address for HTTP proxy location.
<VRF-NAME>	Specifies VRF for HTTP proxy.

A FQDN or IPv4 address are optional in the **no** form of the command.

Examples

Specifying a FQDN for HTTP proxy location and **MGMT** VRF with an IPv6 address:

```
switch(config)# http-proxy http-proxy.example.com vrf mgmt
switch(config)# http-proxy [2000::100]:8080 vrf mgmt
```

Specifying a FQDN for HTTP proxy location and **MGMT** VRF with an IPv4 address:

```
switch(config)# http-proxy http-proxy.example.com vrf mgmt
switch(config)# http-proxy 192.168.1.3:8080 vrf mgmt
```

Removing HTTP proxy location

```
switch(config)# no http-proxy
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

IP source lockdown provides added security by preventing IP source address spoofing on a per-port basis. Every packet is inspected for this purpose in hardware. When IP source lockdown is enabled, IP traffic received on an interface (port) is forwarded only if the VLAN, IP address, MAC address, interface (port) matches the IP binding database entry.

It is best to configure IP source lockdown during a switch maintenance period as enabling it may cause client traffic to be dropped for 10 to 15 seconds.

To use IPv4 source lockdown, the IPv4 binding database must be populated. The binding database is typically dynamically populated by DHCPv4 snooping that learns and saves the binding information. Alternatively, the IPv4 binding database can be statically populated with the `ipv4 source-binding` command described in this chapter. Often DHCPv4 snooping is used to dynamically populate most of the IP binding database along with the `ipv4 source-binding` command that is used to add the binding information for several known and trusted clients, typically administrators. For dynamic IP binding database population with DHCPv4 snooping, see [DHCP snooping](#).

To use IPv6 source lockdown, the IPv6 binding database must be populated. The binding database is typically dynamically populated by DHCPv6 snooping that learns and saves the binding information. Alternatively, the IPv6 binding database can be statically populated with the `ipv6 source-binding` command described in this chapter. Often DHCPv6 snooping is used to dynamically populate most of the IPv6 binding database along with the `ipv6 source-binding` command that is used to add the binding information for several known and trusted clients, typically administrators. For dynamic IPv6 binding database population with DHCPv6 snooping, see [DHCP snooping](#).

IP source lockdown should not be configured on ISL (inter-switch link) ports.

IPv4 source lockdown commands

ipv4 source-binding

```
ipv4 source-binding <VLAN-ID> <IPV4-ADDR> <MAC-ADDR> <IFNAME>
no ipv4 source-binding <VLAN-ID> <IPV4-ADDR> <MAC-ADDR> <IFNAME>
```

Description

Adds static IPv4 client source binding information to the switch IP binding database. Although DHCPv4 snooping is often used to dynamically populate the binding database, this command is available for manually adding entries to the switch IP binding database.

Statically configured IP binding information supersedes any dynamically collected binding information for the same client.

The no form of this command removes the specified binding that was statically configured with the **ipv4 source-binding** command. The no form has no effect on bindings that were dynamically configured with DHCPv4 snooping.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPv4-ADDR>	Specifies the client IPv4 unicast address.
<MAC-ADDR>	Specifies the client MAC address.
<IFNAME>	Specifies the interface on which the client is connected.

Examples

Adding a static IPv4 binding:

```
switch(config)# ipv4 source-binding 1 10.2.1.4 00:50:56:96:e4:cf 1/1/1
```

Removing a IPv4 binding:

```
switch(config)# no ipv4 source-binding 1 10.2.1.4 00:50:56:96:e4:cf 1/1/1
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ipv4 source-lockdown

```
ipv4 source-lockdown  
no ipv4 source-lockdown
```

Description

Enables IPv4 source lockdown for all VLANs on the selected interface (port).

The no form of this command disables IPv4 source lockdown for the selected interface (port).

This configuration will disable flow tracking statistics collection.

Examples

Enabling IPv4 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv4 source-lockdown
```

Enabling IPv4 source lockdown on interface lag112:

```
switch(config)# interface lag112
switch(config-if)# ipv4 source-lockdown
```

Disabling IPv4 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no ipv4 source-lockdown
```

Command History

Release	Modification
10.14	Added information related to role based IPFIX.
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

ipv4 source-lockdown hardware retry

`ipv4 source-lockdown hardware retry <VLAN-ID> <IPV4-ADDR>`

Description

Retries the IPv4 source lockdown hardware programming for a client identified by VLAN and IPv4 address.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPv4-ADDR>	Specifies the client IPv4 unicast address.

Example

Configure IPv4 source lockdown hardware retry for the client on VLAN 10.

```
switch(config)# ipv4 source-lockdown hardware retry 10 1.1.2.1
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show ipv4 source-binding

```
show ipv4 source-binding
```

Description

Shows all IPv4 static source binding information irrespective of source lockdown configuration..

Examples

Showing all IPv4 source binding information:

```
switch# show ipv4 source-binding
```

PORT	VLAN	MAC-ADDRESS	HW-STATUS	FROM	IPv4-ADDRESS
1/1/1	2	aa:bb:cc:dd:ee:ff	Yes	static	1.2.3.4
1/1/2	12	aa:ab:cc:dd:ee:ff	Yes	static	10.20.30.40

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show ipv4 source-lockdown

```
show ipv4 source-lockdown [binding [interface <IFNAME> | ip <IPV4-ADDR> | mac <MAC-ADDR> | vlan <VLAN-ID>] | interface <IFNAME>]
```

Description

Shows summary or detailed IPv4 source lockdown information. When entered without parameters, summary status information for all interfaces (ports) in the binding database is shown.

Parameter	Description
binding	Specifies that detailed lockdown binding record information is to be displayed. The binding database record can be identified by any one of interface (port), ip , mac , or vlan .
interface <IFNAME>	Specifies the client interface (port). When entered without the binding parameter, the summary status information is displayed for the specified interface.
ip <IPV4-ADDR>	Specifies the client IPv4 unicast address.
mac <MAC-ADDR>	Specifies the client MAC address.
vlan <VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

Examples

Showing the summary status information for all interfaces in the binding database:

```
switch# show ipv4 source-lockdown

INTERFACE  LOCKDOWN  HW-STATUS
-----
1/1/1      Yes       Yes
1/1/2      Yes       No
lag112     Yes       Yes
```

Showing the summary status information for the specified interface in the binding database:

```
switch# show ipv4 source-lockdown interface 1/1/2
```

INTERFACE	LOCKDOWN	HW-STATUS
-----	-----	-----
1/1/2	Yes	No

Showing the detailed binding record and related information for all interfaces in the binding database:

```
switch# show ipv4 source-lockdown binding
```

```

Interface Name      : 1/1/1
VLAN Id             : 2000
MAC Address         : 00:50:56:96:e4:cf
IP Address          : 192.168.142.113
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --

Interface Name      : 1/1/2
VLAN Id             : 100
MAC Address         : 00:50:56:96:04:4d
IP Address          : 120.168.43.52
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable

Interface Name      : lag112
VLAN Id             : 12
MAC Address         : 00:50:56:96:d8:3d
IP Address          : 120.168.76.182
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --

Interface Name      : 1/1/1
VLAN Id             : 2000
MAC Address         : 00:50:56:96:e4:cf
IP Address          : 192.168.142.113
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --

Interface Name      : 1/1/2
VLAN Id             : 100
MAC Address         : 00:50:56:96:04:4d
IP Address          : 120.168.43.52
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable

Interface Name      : lag112
VLAN Id             : 12
MAC Address         : 00:50:56:96:d8:3d
IP Address          : 120.168.76.182
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --

```

Showing the detailed binding record and related information for interface 1/1/2:

```
switch# show ipv4 source-lockdown binding interface 1/1/2

Interface Name      : 1/1/2
VLAN Id            : 100
MAC Address         : 00:50:56:96:04:4d
IP Address          : 120.168.43.52
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable

Interface Name      : 1/1/2
VLAN Id            : 100
MAC Address         : 00:50:56:96:04:4d
IP Address          : 120.168.43.52
Time Remaining      : 115 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable
```

Showing the detailed binding record and related information for interface lag112 (identified in this example command by the IP address):

```
switch# show ipv4 source-lockdown binding ip 120.168.76.182

Interface Name      : lag112
VLAN Id            : 12
MAC Address         : 00:50:56:96:d8:3d
IP Address          : 120.168.76.182
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/1 (identified in this example command by the MAC address):

```
switch# show ipv4 source-lockdown binding mac 00:50:56:96:e4:cf

Interface Name      : 1/1/1
VLAN Id            : 2000
MAC Address         : 00:50:56:96:e4:cf
IP Address          : 192.168.142.113
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/2 (identified in this example command by the VLAN):

```
switch# show ipv4 source-lockdown binding vlan 100

Interface Name      : 1/1/2
```

```
VLAN Id           : 100
MAC Address       : 00:50:56:96:04:4d
IP Address        : 120.168.43.52
Time Remaining    : 115 seconds
Lockdown Status   : Yes
Hardware Status    : No
Hardware Error Reason : Resource unavailable
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

IPv6 source lockdown commands

ipv6 source-binding

```
ipv6 source-binding <VLAN-ID> <IPv6-ADDR> <MAC-ADDR> <IFNAME>
no ipv6 source-binding <VLAN-ID> <IPv6-ADDR> <MAC-ADDR> <IFNAME>
```

Description

Adds static IPv6 client source binding information to the switch IPv6 binding database. Although DHCPv6 snooping is often used to dynamically populate the binding database, this command is available for manually adding entries to the switch IPv6 binding database.

Statically configured IPv6 binding information supersedes any dynamically collected binding information for the same client.

The no form of this command removes the specified binding that was statically configured with the **ipv6 source-binding** command. The no form has no effect on bindings that were dynamically configured with DHCPv6 snooping.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPv6-ADDR>	Specifies the client IPv6 address.

Parameter	Description
<MAC-ADDR>	Specifies the client MAC address.
<IFNAME>	Specifies the interface on which the client is connected.

Examples

Adding a static IPv6 binding:

```
switch(config)# ipv6 source-binding 2 2000::2 00:12:11:44:55:12 1/1/28
```

Removing a IPv6 binding:

```
switch(config)# no ipv6 source-binding 2 2000::2 00:12:11:44:55:12 1/1/28
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

ipv6 source-lockdown

```
ipv6 source-lockdown
no ipv6 source-lockdown
```

Description

Enables IPv6 source lockdown for all VLANs on the selected interface (port).

The no form of this command disables IPv6 source lockdown for the selected interface (port).

This configuration will disable flow tracking statistics collection.

Examples

Enabling IPv6 source lockdown on interface 1/1/1:


```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 source-lockdown
```

Enabling IPv6 source lockdown on interface lag112:

```
switch(config)# interface lag112
switch(config-if)# ipv6 source-lockdown
```

Disabling IPv6 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no ipv6 source-lockdown
```

Command History

Release	Modification
10.14	Added information related to role based IPFIX.
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

ipv6 source-lockdown hardware retry

ipv6 source-lockdown hardware retry <VLAN-ID> <IPv6-ADDR>

Description

Retries the IPv6 source lockdown hardware programming for a client identified by VLAN and IPv6 address.

Parameter	Description
<VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.
<IPv6-ADDR>	Specifies the client IPv6 address.

Example

Configure IPv6 source lockdown hardware retry for the client on VLAN 1.

```
switch(config)# ipv6 source-lockdown hardware retry 1 2000::2
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show ipv6 source-binding

```
show ipv6 source-binding
```

Description

Shows all IPv6 static source binding information irrespective of source lockdown configuration.

Examples

Showing all IPv6 source binding information:

```
switch# show ipv6 source-binding
```

PORT	VLAN	MAC-ADDRESS	HW-STATUS	FROM	IPv6-ADDRESS
-----	-----	-----	-----	-----	-----
1/1/1	1234	00:50:56:96:e4:cf	Yes/No	static	3000::1
1/1/1	1	00:50:56:96:04:4d	Yes/No	static	3000::2
1/1/24	1	00:01:01:00:00:01	Yes	static	1001::1

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show ipv6 source-lockdown

```
show ipv6 source-lockdown [binding [interface <IFNAME> | ip <IPV6-ADDR> | mac <MAC-ADDR>
| vlan <VLAN-ID>] | interface <IFNAME>]
```

Description

Shows summary or detailed IPv6 source lockdown information. When entered without parameters, summary status information for all interfaces (ports) in the binding database is shown.

Parameter	Description
binding	Specifies that detailed lockdown binding record information is to be displayed. The binding database record can be identified by any one of interface (port), ip , mac , or vlan .
interface <IFNAME>	Specifies the client interface (port). When entered without the binding parameter, the summary status information is displayed for the specified interface.
ip <IPV6-ADDR>	Specifies the client IPv6 address.
mac <MAC-ADDR>	Specifies the client MAC address.
vlan <VLAN-ID>	Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

Examples

Showing the summary status information for all interfaces in the binding database:

```
switch# show ipv6 source-lockdown

INTERFACE  LOCKDOWN  HW-STATUS
-----
1/1/1      Yes       Yes
1/1/2      Yes       Yes
lag112     Yes       Yes
```

Showing the summary status information for the specified interface in the binding database:

```
switch# show ipv6 source-lockdown interface 1/1/2

INTERFACE  LOCKDOWN  HW-STATUS
-----
1/1/2      Yes       No
```

Showing the detailed binding record and related information for all interfaces in the binding database:

```
switch# show ipv6 source-lockdown binding

Interface Name      : 1/1/1
VLAN Id             : 1234
MAC Address         : 00:50:56:96:e4:cf
IP Address          : aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining      : static
```

```

Lockdown Status      : Yes
Hardware Status      : Yes
Hardware Error Reason : --

Interface Name       : 1/1/2
VLAN Id              : 1234
MAC Address          : 00:50:56:96:04:4d
IP Address           : 4000::1
Time Remaining       : 3290 seconds
Lockdown Status      : Yes
Hardware Status      : No
Hardware Error Reason : Resource unavailable

Interface Name       : lag112
VLAN Id              : 151
MAC Address          : 00:50:56:96:d8:3d
IP Address           : 1001::5
Time Remaining       : 1200 seconds
Lockdown Status      : No
Hardware Status      : Yes
Hardware Error Reason : --

Interface Name       : 1/1/1
VLAN Id              : 1234
MAC Address          : 00:50:56:96:e4:cf
IP Address           : aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining       : static
Lockdown Status      : Yes
Hardware Status      : Yes
Hardware Error Reason : --

Interface Name       : 1/1/2
VLAN Id              : 1234
MAC Address          : 00:50:56:96:04:4d
IP Address           : 4000::1
Time Remaining       : 3290 seconds
Lockdown Status      : Yes
Hardware Status      : No
Hardware Error Reason : Resource unavailable

Interface Name       : lag112
VLAN Id              : 151
MAC Address          : 00:50:56:96:d8:3d
IP Address           : 1001::5
Time Remaining       : 1200 seconds
Lockdown Status      : No
Hardware Status      : Yes
Hardware Error Reason : --

```

Showing the detailed binding record and related information for interface 1/1/2:

```

switch# show ipv6 source-lockdown binding interface 1/1/2

Interface Name       : 1/1/2
VLAN Id              : 1234
MAC Address          : 00:50:56:96:04:4d
IP Address           : 4000::1
Time Remaining       : 3290 seconds
Lockdown Status      : Yes
Hardware Status      : No
Hardware Error Reason : Resource unavailable

```

```
Interface Name      : 1/1/2
VLAN Id             : 1234
MAC Address         : 00:50:56:96:04:4d
IP Address          : 4000::1
Time Remaining      : 3290 seconds
Lockdown Status     : Yes
Hardware Status     : No
Hardware Error Reason : Resource unavailable
```

Showing the detailed binding record and related information for interface 1/1/2 (identified in this example command by the IP address):

```
switch# show ipv6 source-lockdown binding ip 4000::1

Interface Name      : 1/1/2
VLAN Id             : 1234
MAC Address         : 00:50:56:96:04:4d
IP Address          : 4000::1
Time Remaining      : 515 seconds
Lockdown Status     : No
Hardware Status     : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/1 (identified in this example command by the MAC address):

```
switch# show ipv6 source-lockdown binding mac 00:50:56:96:e4:cf

Interface Name      : 1/1/1
VLAN Id             : 1234
MAC Address         : 00:50:56:96:e4:cf
IP Address          : aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining      : static
Lockdown Status     : Yes
Hardware Status     : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface lag112 (identified in this example command by the VLAN):

```
switch# show ipv6 source-lockdown binding vlan 151

Interface Name      : lag112
VLAN Id             : 151
MAC Address         : 00:50:56:96:d8:3d
IP Address          : 1001::5
Time Remaining      : 1200 seconds
Lockdown Status     : No
Hardware Status     : Yes
Hardware Error Reason : --
```

Command History

Release	Modification
10.10	Command enabled on 4100i, 6000 and 6100 series switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

Chapter 9

Internet Control Message Protocol

The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet protocol suite. The protocol is used by network devices, including routers, to send error messages and operational information. For example, an ICMP message might indicate that a requested service is not available. Another example of an ICMP message might be that a host or router could not be reached.

ICMP message types

The type field identifies the type of message sent by the host or gateway.

Type	ICMP messages
0	Echo Reply (Ping Reply, used with Type 8, Ping Request)
3	Destination Unreachable
4	Source Quench
5	Redirect
8	Echo Request (Ping Request, used with Type 0, Ping Reply)
9	Router Advertisement (Used with Type 9)
10	Router Solicitation (Used with Type 10)
11	Time Exceeded
12	Parameter Problem
13	Timestamp Request (Used with Type 14)
14	Timestamp Reply (Used with Type 13)
15	Information Request (obsolete) (Used with Type 16)
16	Information Reply (obsolete) (Used with Type 15)
17	Address Mask Request (Used with Type 17)
18	Address Mask Reply (Used with Type 18)

When ICMP messages are sent

ICMP messages are sent when one or more of the following scenarios occur:

- A datagram cannot reach its destination.
- The gateway does not have the buffering capacity to forward a datagram.
- The gateway can direct the host to send traffic on a shorter route.

ICMP redirect messages

ICMP redirect messages are used by routers to notify the hosts on the data link that a better route is available for a particular destination.

When ICMP redirect messages are sent

The switch is configured to send redirects by default. ICMP redirect messages are sent when one or more of the following scenarios occur:

- The interface on which the packet comes into the router is the same interface on which the packet gets routed out.
- The subnet or network of the source IP address is on the same subnet or network of the next-hop IP address of the routed packet.
- The datagram is not source-routed.
- The destination unicast address is unreachable. In this case, the router generates the ICMP destination unreachable message to inform the source host about the situation.

ICMP commands

ip icmp redirect

```
ip icmp redirect
no ip icmp redirect
```

Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default. The **no** form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

Examples

Enabling ICMP redirect messages:

```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip icmp throttle

```
ip icmp throttle <PACKET-INTERVAL>
no ip icmp throttle [<PACKET-INTERVAL>]
```

Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The **no** form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

Parameter	Description
<PACKET-INTERVAL>	Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 second. Range: 1-86400.

Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

Command History

Release	Modification
10.8	Added the optional <PACKET-INTERVAL> parameter to the no form of the command.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip icmp unreachable

```
ip icmp unreachable
no ip icmp unreachable
```

Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packet. Default setting.

The **no** form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

The Domain Name System (DNS) is the Internet protocol for mapping a hostname to its IP address. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```

switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns

```

```

VRF Name : default
Domain Name: switch.com
Name Server(s): 1.1.1.1

```

Host Name	Address
-----	-----
myhost1	3.3.3.3
switch#	

DNS client commands

ip dns domain-list

```

ip dns domain-list <DOMAIN-NAME> [vrf default]
no ip dns domain-list <DOMAIN-NAME> [vrf default]

```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
<code>list <DOMAIN-NAME></code>	Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```

switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com

```

This example removes the entry **domain1.com**.

```

switch(config)# no ip dns domain-list domain1.com

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns domain-name

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
<DOMAIN-NAME>	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns host

```
ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
host <HOST-NAME>	Specifies the name of a host. Length: 1 to 256 characters.
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns server address

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]  
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

show ip dns

```
show ip dns [vrf <VRF-NAME>]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
<code>vrf <VRF-NAME></code>	Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

Usage

AOS-CX gives the following priority levels to the these three update methods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Examples

These examples define DNS settings and then show how they are displayed with the **show ip dns** command.

```
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host2 2.2.2.2
switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf default
switch(config)# ip dns domain-list reddomain5.com vrf default
switch(config)# ip dns domain-list reddomain8.com vrf default
switch(config)# ip dns server-address 4.4.4.5 vrf default
switch(config)# ip dns server-address 6.6.6.7 vrf default
switch(config)# ip dns host host3 5.5.5.6 vrf default
switch(config)# ip dns host host2 2.2.2.3 vrf default
switch(config)# ip dns host host3 c::13 vrf default
switch# show ip dns
VRF Name : default

Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6

Host Name      Address
-----
host2          2.2.2.2
host3          5.5.5.5
host3          c::12
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	

ARP (Address Resolution Protocol) is used to map the network address assigned to a device to its physical address. For example, on an Ethernet network, ARP maps layer 3 IPv4 network addresses to layer 2 MAC addresses. (ARP does not work with IPv6 addresses. Instead, the Neighbor discovery protocol is used.)

ARP operates at layer 2. ARP requests are broadcast to all devices on the local network segment and are not forwarded by routers. ARP is enabled by default and cannot be disabled.

Proxy ARP

Proxy ARP allows a routing switch to answer ARP requests from devices on one network on behalf of devices on another network. The ARP proxy is aware of the location of the traffic destination, and offers its own MAC address as the final destination.

For example, if Proxy ARP is enabled on a routing switch connected to two subnets (10.10.10.0/24 and 20.20.20.0/24), the routing switch can respond to an ARP request from 10.10.10.69 for the MAC address of the device with IP address 20.20.20.69.

Typically, the host that sent the ARP request then sends its packets to the switch that has the ARP proxy. This switch then forwards the packets to the intended host through a mechanism such as a tunnel.

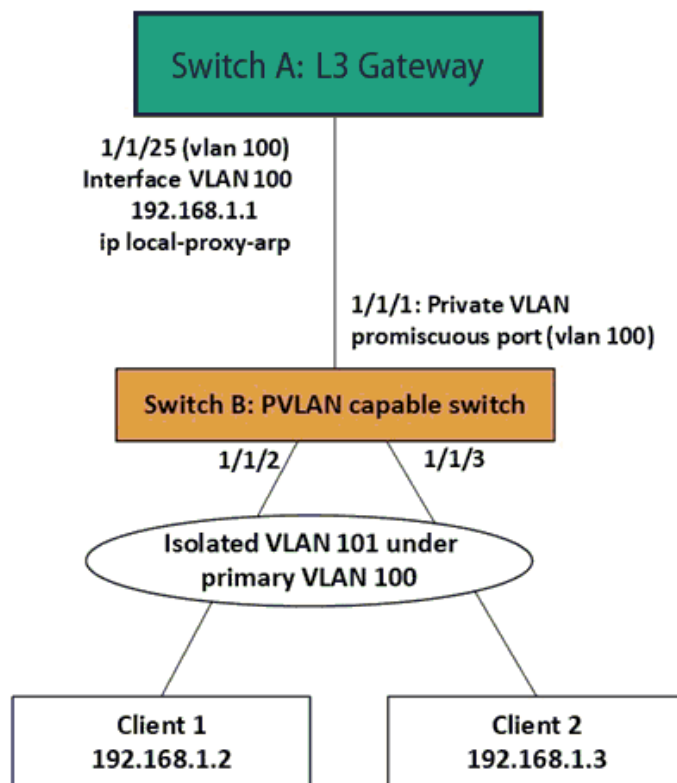
Proxy ARP is supported on L3 physical and VLAN interfaces. It is disabled by default. To enable proxy ARP, routing must be enabled on the interface.

Local proxy ARP

Local proxy ARP is a technique by which a device on a given network answers the ARP queries for a host address that is on the same network. It is primarily used to enable layer 3 communication between hosts within a common subnet that are separated by layer 2 boundaries (Example: PVLAN). Local proxy ARP is supported on L3 physical and VLAN interfaces and is disabled by default. Routing must be enabled on the interface to enable local proxy ARP.

Local proxy ARP can be used along with private VLAN deployments to allow layer 3 communication where clients are isolated at layer 2.

Figure 1 *Local Proxy ARP Use Case Example*



The local proxy ARP feature is to be configured only if there is a need for the clients to communicate while still maintaining layer 2 isolation.

The following steps configure the local ARP proxy deployment shown in [Figure 1, Local Proxy ARP Use Case Example](#).

Step 1: Configuring private VLAN in switch B

- A. Configure the primary and isolated VLAN first

```
PVLAN-Switch(config)# vlan 100
PVLAN-Switch(config-vlan-100)# private-vlan primary
PVLAN-Switch(config-vlan-100)# vlan 101
PVLAN-Switch(config-vlan-101)# private-vlan isolated primary-vlan 100
```

- B. Configure the PVLAN secondary ports to which clients are connected

```
PVLAN-Switch(config)# interface 1/1/2,1/1/3
PVLAN-Switch(config-if-<1/1/1,1/1/2>)# vlan trunk allowed 101
PVLAN-Switch(config-if-<1/1/1,1/1/2>)# private-vlan port-type secondary
```

- C. Configure the PVLAN promiscuous port on the uplink which connects to the L3 gateway

```
PVLAN-Switch(config)# interface 1/1/1
PVLAN-Switch(config-if)# vlan trunk allowed 100
```

```
PVLAN-Switch(config-if) # private-vlan port-type promiscuous
PVLAN-Switch(config-if) # no shutdown
```

For details about private VLAN configuration, refer to the *AOS-CX Layer-2 Bridging guide*.

Step 2: Configuring L3 gateway in switch A

```
L3-GW(config) # vlan 100
L3-GW(config-vlan-100) # exit
L3-GW(config) # interface 1/1/25
L3-GW(config-if) # vlan trunk allowed 100
L3-GW(config-if) # no shutdown
L3-GW(config-if) # exit
L3-GW(config) # interface vlan 100
L3-GW(config-if-vlan) # ip address 192.168.1.1/24
```

Now the clients will be able to ping the gateway IP; but will not be able to communicate with each other due to the isolation enforced with private VLAN. As expected, the ping from client 1 to client 2 will fail in the above example, although the clients are in the same subnet.

Step 3: Configuring local proxy ARP in switch A

If local proxy ARP is enabled in the L3 gateway, the L3 gateway responds to ARP requests for addresses in the same subnet. In other words, the L3 gateway will be doing proxy for ARP resolution.

```
L3-GW(config) # no ip icmp redirect
L3-GW(config) # interface vlan 100
L3-GW(config-if-vlan) # ip local-proxy-arp
```

With this, the L3 communication will be enabled between client 1 and client 2. Now the ping from client 1 to client 2 will be successful in the above example. The ARP table in client 1 will be populated with the MAC of L3 gateway; not the MAC of client 2. Similar ARP table entry for client 2 as well. In other words, the L3 gateway is doing proxy for ARP resolution for the local network where clients are isolated at layer 2.

Dynamic ARP inspection

Dynamic arp inspection is provided as a mechanism for making ARP more secure.

ARP is used for resolving IP against MAC addresses on a broadcast network segment like the Ethernet and was originally defined by Internet Standard RFC 826. ARP does not support any inherent security mechanism and as such depends on simple datagram exchanges for the resolution, with many of these being broadcast.

Because it is an unreliable and non-secure protocol, ARP is vulnerable to attacks. Some attacks may be targeted toward the networks whereas other attacks may be targeted toward the switch itself. The attacks primarily intend to create denial of service (DoS) for the other entities present in the network.

Most of the attacks are carried out in one of the following three forms:

- Overwhelming the switch control plane with too many ARP packets.
- Overwhelming the switch control plane with too many unresolved data packets.
- Masquerading as a trusted gateway/server by wrongly advertising ARPs.

Several defense mechanisms can be put in place on a switch to protect against attacks:

- Limit the amount of ARP activity allowed from a host or on a port.
- Ensure that all ARP packets are consistent with one or more binding databases, which can be created through various means.
- Enforce integrity checks on the ARP packets to check against different MAC or IP addresses in the Ethernet or IP header and ARP header.

The following is supported:

- Enabling and disabling of Dynamic ARP Inspection on a VLAN level (it does not have to be SVI).
- Defining the member ports of a VLAN as either trusted or untrusted.
- Only ARP traffic on untrusted ports subjected to checks.
- Routed ports (RoPs) always treated as trusted.
- Listening to the DHCP Bindings table and check every ARP packet to match against the binding.

ARP ACLs are not supported and the DHCP snooping table is the only source of binding.

ARP commands

arp inspection

arp inspection

Description

Enables Dynamic ARP inspection on the current VLAN, which means that ARP packets received from untrusted interfaces are discarded if they have an Invalid IP-to-MAC address binding.

The **no** form of this command disables Dynamic ARP Inspection on the VLAN.

Examples

Enabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# arp inspection
```

Disabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# no arp inspection
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-vlan-<VLAN-ID>	Administrators or local user group members with execution rights for this command.

arp inspection trust

```
arp inspection trust
no arp inspection trust
```

Description

Configures the interface as a trusted. All interfaces are untrusted by default. The **no** form of this command returns the interface to the default state (untrusted).

Example

Setting an interface as trusted:

```
switch(config-if) # arp inspection trust
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

arp ip

```
arp ip <IP_ADDR> mac <MAC_ADDR>
no arp ip <IP_ADDR> mac <MAC_ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors). The no form of this command deletes a permanent static neighbor entry from the ARP table.

Parameter	Description
ip <IP-ADDR>	Specifies the IP address of the neighbor or the virtual IP address

Parameter	Description
	of the cluster in IP format (x.x.x.x), where x is a decimal number from 0 to 255. . Range: 4096 to 131072. Default: 131072.
mac <MAC-ADDR>	Specifies the MAC address of the neighbor or the multicast MAC address in IANA format (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Example

Configuring a static ARP entry on a interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# arp ip 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no arp ip 2.2.2.2 mac 01:00:5e:00:00:01
```

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan	Administrators or local user group members with execution rights for this command.

arp process-grat-arp

```
arp process-grat-arp
no arp process-grat-arp
```

Description

Enables the processing of gratuitous ARP packets on the individual port or group of L3 ports together. By default, the gratuitous ARP processing is enabled. When gratuitous ARP (GARP) processing is enabled, a switch that is advertising any changes in its MAC through the GARP will reflect in the neighbor table of the switch. However, the switch will not be able to learn the neighbor through the GARP. This configuration is applicable only on L3 interfaces such as ROPs, subinterfaces, and SVIs. The **no** form of this command disables the processing of gratuitous ARP packets.

Example

Enabling the processing of gratuitous ARP packets on the interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# arp process-grat-arp
```

Enabling the processing of gratuitous ARP packets on interfaces **1/1/1** to **1/1/5**:

```
switch(config)# interface 1/1/1-1/1/5
switch(config-if<1/1/1-1/1/5>)# no shutdown
switch(config-if<1/1/1-1/1/5>)# arp process-grat-arp
```

Enabling the processing of gratuitous ARP packets on sub-interface **1/1/1.10**:

Applies only to the HPE Aruba Networking 6300, 6400, 8100, and 8360 Switch Series.

```
switch(config)# interface 1/1/1.10
switch(config-subif)# no shutdown
switch(config-subif)# arp process-grat-arp
```

Disabling the processing of gratuitous ARP packets on VLANs **2** to **100**:

```
switch(config)# interface vlan 2-100
switch(config-if-vlan<2-100>)# no shutdown
switch(config-if-vlan<2-100>)# no arp process-grat-arp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan config-subif	Administrators or local user group members with execution rights for this command.

clear arp

```
clear arp
port <PORT-ID> [ip <A.B.C.D>|all] [[ipv6 <X:X::X:X>|all]vla
vrf [all-vrfs|{<VRF-NAME> [ip <A.B.C.D>] [[ipv6 <X:X::X:X>]]}]
```

Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any VRF or port parameters, ARP table entries are cleared for the **default** VRF.

Parameter	Description
port <PORT-ID>	Specifies a port on the switch. For example: vlan 2 .
ip <A.B.C.D> all]	(Optional) Include an IP address to clear neighbor entries for that specific address, or use the all parameter to clear entries for all IP addresses.
ipv6 <X:X::X:X> all	(Optional) Include an IPv6 address to clear neighbor entries for that specific address, or use the all parameter to clear entries for all IPv6 addresses.
vrf	Clears IPv4 and IPv6 neighbor entries for the specified VRF or for all VRFs. If no VRF is specified the default VRF is cleared.
all-vrfs	Clear neighbor entries for all VRFs
<VRF-NAME>	Clear neighbor entries for the specified VRF.
ip <A.B.C.D>	(Optional) Include an IP address to clear just the neighbor entries for the specified IP address.
ipv6 <X:X::X:	(Optional) Include an IPv6 address to clear the neighbor entries for the specified IPv6 address.

Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:

```
switch# clear arp
```

Clearing all ARP neighbor entries for a port:

```
switch# clear arp vlan 2
```

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

debug arp-security

```
debug arp-security <LOG-CATEGORY> [severity <LEVEL>]
no debug arp-security [<LOG-CATEGORY>] [severity <LEVEL>]
```

Description

Enables ARP security debug logs. If <SEVERITY> is omitted, all severities are logged.

The **no** form of this command disables ARP security debug logs.

Parameter	Description
<code><LOG-CATEGORY></code>	Selects the ARP security debug log category. Available categories are: ▪ all: Selects all ARP security debug log categories. ▪ config: Selects the ARP security config debug log category. ▪ inspection: Selects the ARP security inspection debug log category. ▪ packet: Selects the ARP security packet debug log category.
<code>severity <LEVEL></code>	Specifies how to filter the ARP security debug logging by setting the minimum severity level for which debug logging will be performed. The selected severity level and all severities above (more severe) will be included in the logging. ▪ emerg: Sets ARP security debug log filtering to Emergency only. ▪ alert: Sets ARP security debug log filtering to Alert and above. ▪ critical: Sets ARP security debug log filtering to Critical and above. ▪ error: Sets ARP security debug log filtering to Error and above. ▪ warning: Sets ARP security debug log filtering to Warning and above. ▪ notice: Sets ARP security debug log filtering to Notice and above. ▪ info: Sets ARP security debug log filtering to Info and above. ▪ debug: Sets ARP security debug log filtering to all severities.

Examples

Enable ARP security debug logging for all categories and all severities:

```
switch# debug arp-security all
```

Enable ARP security config debug log for severity level Error and above:

```
switch# debug arp-security config severity error
```

Enable ARP security inspection debug log for severity level Notice and above:

```
switch# debug arp-security inspection severity notice
```

Enable ARP security debug packet for severity level Critical and above:

```
switch# debug arp-security packet severity critical
```

Enable ARP security debug logging for all categories and severity level Alert and above:

```
switch# debug arp-security all severity alert
```

Disable ARP security debug logging:

```
switch# no debug arp-security
```

Command History

Release	Modification
10.11	Command introduced on 4100i Switch Series
10.09	Command introduced on 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

ip proxy-arp

```
ip proxy-arp  
no ip proxy-arp
```

Description

Enables proxy ARP for the specified Layer 3 interface. Proxy ARP is supported on Layer 3 physical interfaces, LAG interfaces, and VLAN interfaces. It is disabled by default. To enable proxy ARP on an interface, routing must be enabled on that interface.

The **no** form of this command disables proxy ARP for the specified interface.

Examples

Enabling proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1  
switch(config-if) # ip proxy-arp
```

Enabling proxy ARP on VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip proxy-arp
```

Enabling proxy ARP on a LAG 11:

```
switch(config)# int lag 11
switch(config-lag-if)# ip proxy-arp
```

Disabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# no ip proxy-arp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if config-if-vlan config-lag-vlan	Administrators or local user group members with execution rights for this command.

ipv6 neighbor mac

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The **no** form of this command deletes a permanent static neighbor entry from the ARP table.

Parameter	Description
<IPV6-ADDR>>	Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.
mac <MAC-ADDR>>	Specifies the MAC address of the neighbor (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Example

Creates a static ARP entry on interface **vlan 2**.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334 mac
00:50:56:96:df:c8
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

show arp

show arp

Description

Shows the entries in the ARP (Address Resolution Protocol) table.

Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

Examples

```
switch# show arp

IPv4 Address      MAC                Port      Physical Port
-----
192.168.1.2       00:50:56:96:7b:e0  vlan10    1/1/29      stale
192.168.1.3       00:50:56:96:7b:ac  vlan10    1/1/1       reachable

Total Number Of ARP Entries Listed- 2.
-----
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show arp inspection interface

show arp inspection interface [<IFNAME>] [vlan <VLAN-ID>]

Description

Shows the current configuration of dynamic ARP inspection on an interface.

Parameter	Description
<IFNAME>	Specifies the interface.
<VLAN-ID>	Specifies the VLAN ID. Range: 1 to 4094.

Examples

Showing current configuration of dynamic ARP inspection on all interfaces:

```
switch# show arp inspection interface
```

```
-----
Interface          Trust-State
-----
1/1/1              Untrusted
-----
```

Showing current configuration of dynamic ARP inspection on a particular interface:

```
switch# show arp inspection interface 1/1/1
```

```
-----
Interface          Trust-State
-----
1/1/1              Untrusted
-----
```

Showing current configuration of dynamic ARP inspection on interface VLAN 2:

```
switch# show arp inspection interface vlan 2
```

```
-----
Interface          Trust-State
-----
vlan2              Trusted
-----
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show arp inspection statistics

show arp inspection statistics vlan [<VLAN-ID>]

Description

Shows statistics about forwarded and dropped ARP packets. When <VLAN-ID> is not specified, information is shown for all configured VLANs.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID or range of IDs separated by a dash "-". Range: 1 to 4094.

Examples

Showing ARP packet statistics for a range of VLANs:

```
switch# show arp inspection statistics vlan 1-100
```

```
-----
VLAN   Name                Forwarded    Dropped
-----
1      DEFAULT_VLAN_1      0           0
-----
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i	Operator (>) or Manager	

Platforms	Command context	Authority
6000 6100	(#)	

show arp inspection vlan

show arp inspection vlan [<VLAN-ID>]

Description

Shows the current configuration of dynamic ARP inspection on a VLAN. When <VLAN-ID> is not specified, information is shown for all configured VLANs.

Parameter	Description
<VLAN-ID>	Specifies the VLAN ID or range of IDs separated by a dash "-". Range: 1 to 4094.

Examples

Showing dynamic ARP configuration for all VLANs:

```
switch# show arp inspection vlan
```

```
-----
VLAN    Name                ARP Inspection
-----
1       DEFAULT_VLAN_1    -
100     VLAN100            -
200     VLAN200            Enabled
-----
```

Showing dynamic ARP configuration for a particular VLAN:

```
switch# show arp inspection vlan 1
```

```
-----
VLAN    Name                ARP Inspection
-----
1       DEFAULT_VLAN_1    -
-----
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	

show arp state

```
show arp state {all | failed | incomplete | permanent | reachable | stale}
```

Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

Parameter	Description
all	Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.
failed	Shows the ARP cache entries that are in <code>failed</code> state. The neighbor might have been deleted.
incomplete	Shows the ARP cache entries that are in <code>incomplete</code> state. An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout.
permanent	Shows the ARP cache entries that are in <code>permanent</code> state. ARP entries that are in a permanent state can be removed by administrative action only.
reachable	Shows the ARP cache entries that are in <code>reachable</code> state, meaning that the neighbor is known to have been reachable recently.
stale	Shows ARP cache entries that are in <code>stale</code> state. ARP cache entries are in the <code>stale</code> state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Examples

```
switch# show arp state failed
```

IPv4 Address	MAC	Port	Physical Port	State
192.168.1.4		vlan10		failed

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show arp summary

show arp summary [all-vrfs | vrf <VRF-NAME>]

Description

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

Parameter	Description
all-vrfs	Selects all VRFs.
vrf <VRF-NAME>	Specifies the name of a VRF.

Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs

ARP Entry's State           : IPv4      IPv6
-----
Number of Reachable ARP entries : 2        0
Number of Stale ARP entries   : 0        0
Number of Failed ARP entries  : 2        2
Number of Incomplete ARP entries : 0        0
Number of Permanent ARP entries : 0        0
-----
Total ARP Entries: 6         : 4        2
-----
```

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```
vsx-primary# show arp summary

ARP Entry's State           IPv4      IPv6
-----
Number of Reachable ARP entries 25858    32231
Number of Stale ARP entries      0         1
Number of Failed ARP entries     0       257
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries  0         0
-----
```

```
Total ARP Entries- 58347          25858          32489
```

```
vsx-primary# show arp summary vsx-peer
```

```
ARP Entry's State          IPv4          IPv6
-----
Number of Reachable ARP entries  25858        32168
Number of Stale ARP entries      0             3
Number of Failed ARP entries     0            317
Number of Incomplete ARP entries 0             0
Number of Permanent ARP entries  0             0
-----
Total ARP Entries- 58346          25858          32488
-----
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show arp timeout

```
show arp timeout [<INTERFACE>]
```

Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

Parameter	Description
<INTERFACE>	Specifies a physical port, VLAN, or LAG on the switch. For physical ports, use the format (for example, 1/3/1).

Examples

Showing ARP timeout information for a VLAN:

```
switch# show arp timeout vlan2
Port          VRF          Timeout
-----
vlan2         default      1800
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show arp vrf

```
show arp {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows the ARP table for all VRF instances, or for the named VRF.

Parameter	Description
all-vrfs	Specifies all VRFs.
vrf <VRF-NAME>	Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

Examples

Showing ARP entries for VRF **vrf1**.

```
switch# show arp vrf vrf1

IPv4 Address      MAC                Port      Physical Port      State
VRF
-----
100.1.250.50      00:50:56:8d:44:13  vlan1001   1/1/2
reachable vrf1
100.2.250.60      00:50:56:8d:45:63  vlan1002   vxlan1(1920:1680:1:1::2)
permanent vrf1

Total Number Of ARP Entries Listed: 2.
-----
```

This example from a different network shows ARP entries for all VRFs.

```
switch# show arp all-vrfs
ARP IPv4 Entries:
-----
IPv4 Address      MAC                Port      Physical Port      State      VRF
192.168.120.10    00:50:56:bd:10:be  1/1/32    1/1/32             reachable  red
10.20.30.40       00:50:56:bd:6a:c5  1/1/29    1/1/29             reachable  test
-----
Total Number Of ARP Entries Listed: 2.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show ipv6 neighbors

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

Parameter	Description
<code>all-vrfs</code>	Specifies all VRFs.
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

Examples

```
switch# show ipv6 neighbors
IPv6 Entries:
```

```
-----

IPv6 Address          MAC                Port    Physical Port  State
fe80::a21d:48ff:fe8f:2700  a0:1d:48:8f:27:00  vlan2300  1/1/31         reachable
fe80::f603:43ff:fe80:a600  f4:03:43:80:a6:00  vlan2300  1/1/30         reachable
-----
```

```
Total Number Of IPv6 Neighbors Entries Listed: 2.
-----
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 neighbors state

```
show ipv6 neighbors state {all | failed | incomplete | permanent | reachable | stale}
```

Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

Parameter	Description
all	Shows all ARP cache entries.
failed	Shows ARP cache entries that are in <code>failed</code> state. The neighbor might have been deleted. Set the neighbor to be unreachable.
incomplete	Shows ARP cache entries that are in <code>incomplete</code> state. An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, and you are waiting for a solicitation reply or a timeout.
permanent	Shows ARP cache entries that are in <code>permanent</code> state.
reachable	Shows ARP cache entries that are in <code>reachable</code> state, meaning that the neighbor is known to have been reachable recently.
stale	Shows ARP cache entries that are in <code>stale</code> state. ARP cache entries are in the <code>stale</code> state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Example

```
switch# show ipv6 neighbors state all
```

IPv6 Address	MAC	Port	Physical Port	State
100::2	48:0f:cf:af:f1:cc	lag1	lag1	
reachable				
300::3	48:0f:cf:af:33:be	vlan3	1/4/20	
reachable				
fe80::4a0f:cfff:feaf:f1cc	48:0f:cf:af:f1:cc	lag1	lag1	
reachable				

```

200::3                48:0f:cf:af:33:be  1/4/11  1/4/11
reachable
fe80::4a0f:cfff:feaf:33be  48:0f:cf:af:33:be  vlan3    1/4/20
reachable

```

Total Number Of IPv6 Neighbors Entries Listed- 5.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show tech arp-security

show tech arp-security

Description

Shows the output of these three commands:

- **show arp inspection statistics vlan**
- **show arp inspection vlan**
- **show arp inspection interface**

Examples

Showing the output of the three ARP security show commands:

```

switch(config-if)# show tech arp-security
=====
Show Tech executed on Mon Nov 28  09:53:54 2019
=====
[Begin] Feature arp-security
=====

*****
Command : show arp inspection statistics vlan
*****

-----
VLAN    Name                Forwarded    Dropped
-----
1       DEFAULT_VLAN_1      0           0
200     VLAN200             0           0
-----

```

```
*****
Command : show arp inspection vlan
*****
```

```
-----
VLAN      Name                ARP-Inspection
-----
1         DEFAULT_VLAN_1    -
200      VLAN200        Enabled
-----
```

```
*****
Command : show arp inspection interface
*****
```

```
-----
Interface      Trust-State
-----
1/1/1          Untrusted
lag100         Trusted
-----
```

```
=====
[End] Feature arp-security
=====
```

```
=====
Show Tech commands executed successfully
=====
```

Command History

Release	Modification
10.11	Command introduced on the 4100i Switch Series
10.09	Command introduced on the 6000, 6100 Switch Series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html
AOS-CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

HPE Aruba Networking Hardware Documentation and Translations Portal	https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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(docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.